

Single-Cell Foundation Models for Cell Type Annotation and Perturbation Prediction: A Survey

Single-cell foundation models aim to learn reusable representations from large cell atlases, single-cell transcriptomes, perturbation screens, and multimodal omics data. They promise better cell type annotation, reference mapping, biological state discovery, and perturbation prediction, but they also inherit challenges from sparse counts, batch effects, evolving taxonomies, and limited experimental labels. This survey synthesizes an arXiv-derived corpus of 276 papers collected during this run and organizes the literature around single-cell pretraining, gene and cell representations, cell type annotation, perturbation and drug-response prediction, multimodal and spatial omics, atlas integration, biological discovery, and evaluation. The survey argues that useful single-cell foundation models must be biologically grounded, batch-aware, interpretable, and validated on transfer settings that mirror real experimental use.

Keywords:

single-cell foundation models; cell type annotation; perturbation prediction; gene expression; cell atlas; scRNA-seq; multimodal omics; biological representation learning

1 Introduction

Single-cell technologies measure gene expression, chromatin accessibility, protein abundance, spatial location, and perturbation responses at cellular resolution. These measurements are transforming biology, but they create a modeling problem: cells are sparse, noisy, heterogeneous, and embedded in biological hierarchies that change across tissues, diseases, donors, protocols, and time. Foundation models seek to learn reusable cellular representations from large collections of single-cell data and then transfer them to tasks such as cell type annotation and perturbation prediction [23, 45, 53, 140, 188, 203, 206, 248].

Cell type annotation maps a measured cell to a biological label or state, often by transferring knowledge from a reference atlas. Perturbation prediction asks how a cell state changes after a gene knockout, drug treatment, CRISPR intervention, or environmental stimulus. These tasks are linked: annotation establishes cell identity and context, while perturbation prediction models how identity and

state respond to interventions. Single-cell foundation models promise a shared representation for both [18, 23, 45, 53, 140, 188, 203, 248].

This survey reviews single-cell foundation models for cell type annotation and perturbation prediction. We emphasize pretraining objectives, gene and cell tokenization, atlas integration, multimodal omics, transfer evaluation, biological interpretability, and experimental deployment.

2 Background

2.1 Single-Cell Data and Biological Variation

Single-cell RNA sequencing measures sparse transcript counts for thousands of genes across many cells. Related assays measure chromatin accessibility, cell-surface proteins, spatial coordinates, perturbation identities, or lineage relationships. Biological variation includes cell type, cell state, tissue, developmental stage, donor, disease, and experimental per-

turbation. Technical variation includes batch effects, sequencing depth, dropout, platform differences, and annotation inconsistencies [23, 45, 53, 140, 188, 203, 206, 248].

2.2 Foundation Models in Single-Cell Biology

Foundation models are trained on broad data with objectives that can transfer to many tasks. In single-cell biology, this usually means learning from gene expression matrices, ranked gene lists, tokenized genes, cell atlases, multimodal assays, or perturbation screens. Models may use transformers, autoencoders, contrastive learning, masked gene modeling, graph neural networks, or generative latent-variable models. Their value depends on whether learned representations transfer across tissues, protocols, and tasks rather than only fitting a benchmark [18, 23, 45, 53, 140, 188, 203, 248].

2.3 Annotation and Perturbation as Transfer Tasks

Cell type annotation is a supervised or semi-supervised transfer problem: labels from a reference are mapped to a query dataset. Perturbation prediction is a counterfactual transfer problem: a model predicts a cell state under an unobserved intervention. Both tasks require representations that preserve biology while controlling nuisance variation. Both also need uncertainty, because wrong labels or overconfident perturbation forecasts can mislead downstream experiments [29, 45, 100, 139, 164, 203, 205, 251].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with single-cell foundation models, scGPT, Geneformer, scBERT, scFoundation, universal cell embeddings, cell type annotation, label transfer, reference mapping, perturbation prediction, Perturb-seq, CRISPR screens, virtual perturbation, drug response, gene expression transformers, masked gene modeling, single-cell representation learning, scVI, totalVI, multimodal single-cell omics, scRNA-seq, scATAC-seq, spatial transcriptomics, batch correction, atlas integration, transfer learning, and benchmarks. Records were retained

when title or abstract evidence connected single-cell data with foundation/pretrained models, representation learning, cell annotation, perturbation prediction, multimodal omics, atlas mapping, or evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 276 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *input modality*: scRNA-seq, scATAC-seq, CITE-seq, spatial transcriptomics, perturbation screens, lineage data, or multimodal atlases. The second is *tokenization*: genes, expression ranks, binned counts, cell embeddings, pathways, peaks, proteins, or spatial neighborhoods. The third is *model family*: transformer, masked gene model, autoencoder, variational latent model, graph neural network, contrastive encoder, generative model, or multimodal fusion model. The fourth is *pretraining signal*: masked prediction, reconstruction, contrastive alignment, next-state prediction, batch-invariant embedding, or perturbation response. The fifth is *task*: annotation, label transfer, reference mapping, perturbation prediction, drug response, integration, trajectory inference, or gene-regulatory discovery. The sixth is *transfer setting*: cross-batch, cross-tissue, cross-species, cross-assay, few-shot, zero-shot, or out-of-distribution. The seventh is *evaluation*: accuracy, calibration, biological plausibility, gene-level interpretability, robustness, and experimental utility. The eighth is *deployment*: atlas maintenance, privacy, scalability, reproducibility, and human expert review.

3.3 Pretraining and Foundation Model Design

Single-cell pretraining adapts language-model ideas to biological matrices. Genes can be tokens, expression values can be bins or ranks, and cells can be sequences or sets. Masked gene modeling, contrastive learning, and reconstruction train models to infer expression structure, while atlas-scale training exposes them to many cell states. A central question is which biological invariances should be learned and which differences should be preserved [18, 23, 45, 53, 140, 188, 203, 248].

3.4 Gene and Cell Representation Learning

Representations support downstream tasks only if they encode useful biology. Gene embeddings can re-

flect co-expression, pathways, regulatory programs, and perturbation effects. Cell embeddings can organize cell types, states, trajectories, and disease contexts. Variational and autoencoder-based models remain important because they explicitly model latent structure and technical noise, while transformer models add flexible attention over genes and contexts [18, 23, 53, 70, 87, 105, 134, 154].

3.5 Cell Type Annotation and Label Transfer

Annotation systems transfer labels from reference atlases to query datasets, often under batch shifts and incomplete taxonomies. Foundation models can improve annotation by encoding broad biological context, but they must handle novel cell states, rare populations, ambiguous labels, and inconsistent marker definitions. Good annotation tools should report uncertainty and allow expert review rather than forcing every cell into a known label [29, 45, 100, 139, 164, 203, 205, 251].

3.6 Perturbation Prediction and Virtual Experiments

Perturbation models predict how gene expression or cell state changes after interventions. Perturb-seq and CRISPR screens provide supervised data, but the combinatorial space of genes, cell types, doses, and contexts is vast. Foundation models can share information across genes and cell states, enabling zero-shot or few-shot virtual experiments. The hard cases are combinatorial perturbations, context-specific effects, nonlinear pathways, and distribution shifts [23, 45, 53, 140, 167, 205, 206, 248].

3.7 Multimodal and Spatial Single-Cell Modeling

Single-cell foundation models increasingly combine expression with chromatin, proteins, spatial location, lineage, and metadata. Multimodal learning can align assays and reveal regulatory mechanisms, while spatial data adds tissue context. The challenge is missing modalities: many datasets measure only one view, so models must learn useful cross-modal alignment without over-imputing unsupported biology [18, 23, 45, 139, 167, 175, 203, 251].

3.8 Atlas Integration and Transfer Learning

Large cell atlases motivate reference mapping, batch correction, and harmonized embeddings. Transfer

learning asks whether a model trained on one atlas can annotate or predict in another tissue, disease, species, or protocol. Strong systems must separate biological variation from technical artifacts while preserving rare states and newly discovered cell types [18, 44, 45, 164, 167, 203, 205, 251].

3.9 Biological Discovery and Disease Modeling

Beyond prediction, single-cell foundation models are used for disease subtyping, immune profiling, developmental trajectories, drug response, regulatory-network discovery, and biomarker identification. These applications require interpretability at the gene, pathway, and cell-state levels. A model that performs well on labels may still fail to produce biologically actionable mechanisms [140, 146, 166, 167, 188, 206, 248, 251].

3.10 Benchmarks and Evaluation

Evaluation should distinguish interpolation from transfer. Cell annotation benchmarks should test unseen batches, tissues, species, and rare cell types. Perturbation benchmarks should test held-out genes, combinations, cell types, and experimental contexts. Reports should include calibration, uncertainty, data leakage checks, and biological validation rather than only aggregate accuracy [18, 45, 140, 146, 188, 203, 206, 251].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across foundation-model pretraining, representation learning, annotation, perturbation prediction, multimodal omics, atlas integration, biological discovery, and evaluation.

Single-cell foundation models and pretraining The single-cell foundation models and pretraining cluster contains work that develops methods, systems, datasets, or analyses central to this survey [23, 29, 140, 146, 154, 166, 188, 203, 248]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [40, 41, 56, 70, 83, 102, 106, 108, 164]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [30, 58, 104, 117, 155, 159, 268, 269, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions

[15, 22, 34, 42, 57, 66, 73, 119, 214]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 82, 130, 158, 172, 197, 209, 255, 258]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [26, 50, 165, 213, 235, 265, 276].

Gene and cell representation learning The gene and cell representation learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [5, 18, 53, 98, 105, 182, 194, 259, 267]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [6, 97, 101, 107, 114, 120, 169, 171, 204]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [14, 39, 81, 135, 163, 208, 226].

Cell type annotation and label transfer The cell type annotation and label transfer cluster contains work that develops methods, systems, datasets, or analyses central to this survey [64, 69, 118, 141, 199, 205, 211, 262].

Perturbation prediction and virtual experiments The perturbation prediction and virtual experiments cluster contains work that develops methods, systems, datasets, or analyses central to this survey [44, 63, 78, 109, 134, 167, 181, 190, 256]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [8, 99, 122, 127, 142, 143, 168, 250, 264]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 31, 47, 72, 176, 177, 216, 218, 234]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [28].

Single-cell multimodal and spatial omics The single-cell multimodal and spatial omics cluster contains work that develops methods, systems, datasets, or analyses central to this survey [2, 45, 76, 79, 100, 139, 160, 239, 251]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [13, 25, 71, 94, 133, 147, 189, 223, 232]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 84, 116, 125, 161, 178,

192, 225, 260]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [86, 128, 179, 212, 221, 253, 274, 275].

Batch integration, atlas mapping, and transfer learning The batch integration, atlas mapping, and transfer learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [61, 129, 153, 196].

Biological discovery, disease modeling, and drug response The biological discovery, disease modeling, and drug response cluster contains work that develops methods, systems, datasets, or analyses central to this survey [36, 62, 136, 152, 175, 193, 202, 257, 272]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [46, 95, 115, 132, 173, 198, 217, 244, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 60, 77, 90, 113, 184, 187, 201, 254]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [74, 75, 80, 123, 200, 207, 210, 224, 236]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [43, 68, 126, 149, 156, 185].

Benchmarks, evaluation, interpretability, and deployment The benchmarks, evaluation, interpretability, and deployment cluster contains work that develops methods, systems, datasets, or analyses central to this survey [32, 33, 48, 87, 89, 110, 131, 206, 266]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [20, 51, 91–93, 157, 227, 229, 231]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 112, 151, 183, 186, 233, 243, 249, 263]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 49, 85, 103, 215, 240, 242, 246, 247]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 16, 19, 37, 124, 137, 180, 219, 238]. Additional retained studies broaden the cluster with alternative architectures, training signals,

data sources, and deployment assumptions [54, 65, 88, 121, 145, 148, 220, 252, 261]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 52, 67, 96, 191, 222, 230, 245, 271]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [21, 27, 35, 59, 111, 150, 228, 237, 241]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 17, 38, 55, 138, 144, 162, 170, 195]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [174].

4 Challenges and Prospects

4.1 Biological Grounding of Learned Tokens

Genes are not words and cells are not sentences. Future models need tokenizations and objectives that respect pathways, regulation, copy number, chromatin, spatial context, and experimental design.

4.2 Annotation Under Open-World Cell Taxonomies

Cell atlases evolve. Annotation systems should detect novel cell states, report uncertainty, and support hierarchical labels instead of assuming every query cell belongs to a closed reference taxonomy.

4.3 Counterfactual Validity in Perturbation Prediction

Perturbation prediction is causal in spirit. Models must distinguish correlations from intervention effects and should be evaluated on held-out genes, combinations, and cellular contexts that resemble real experimental planning.

4.4 Multimodal Missingness and Batch Effects

Most atlases are incomplete across modalities and protocols. Foundation models must align data without hallucinating measurements, and benchmarks should expose failure under missing modalities and batch shift.

4.5 Privacy, Governance, and Reproducibility

Single-cell atlases may encode donor, disease, and demographic information. Deployment requires privacy-aware training, clear provenance, reproducible preprocessing, and governance over model updates and reference labels.

5 Conclusion

Single-cell foundation models offer a path toward reusable cellular representations for annotation, perturbation prediction, atlas mapping, multimodal integration, and biological discovery. Their success depends on more than scale: they must handle batch effects, open taxonomies, sparse perturbation labels, multimodal missingness, and biological interpretability. This survey reviewed the literature across pre-training, gene and cell embeddings, annotation, perturbation prediction, multimodal omics, integration, discovery, and evaluation. The central lesson is that foundation models for single-cell biology should be judged by transfer, calibration, and experimental utility, not only by benchmark scores.

References

- [1] Abrar Rahman Abir, Sajib Acharjee Dip, and Liqing Zhang. CFM-GP: Unified Conditional Flow Matching to Learn Gene Perturbation Across Cell Types. *arXiv preprint arXiv:2508.08312*, 2025. URL <https://arxiv.org/abs/2508.08312>.
- [2] Paul H. Acosta, Pingjun Chen, Simon P. Castillo, Maria Esther Salvatierra, Yinyin Yuan, and Xiaoxi Pan. CellSymphony: Deciphering the molecular and phenotypic orchestration of cells with single-cell pathomics. *arXiv preprint arXiv:2508.10232*, 2025. URL <https://arxiv.org/abs/2508.10232>.
- [3] Tejumade Afonja, Ivaxi Sheth, Ruta Binkyte, Waqar Hanif, Thomas Ulas, Matthias Becker, and Mario Fritz. LLM4GRN: Discovering Causal Gene Regulatory Networks with LLMs – Evaluation through Synthetic Data Generation. *arXiv preprint arXiv:2410.15828*, 2024. URL <https://arxiv.org/abs/2410.15828>.
- [4] Alma Andersson, Aya Abdelsalam Ismail, Edward De Brouwer, Doron Haviv, Tommaso Biancalani, Kyunghyun Cho, Gabriele Scalia, Aicha BenTaieb, and Hector Corrada Bravo. scCBGM: Interpretable Single-Cell Counterfactual Editing. *arXiv preprint arXiv:2606.07760*, 2026. URL <https://arxiv.org/abs/2606.07760>.

- [5] Aixa X. Andrade, Son Nguyen, Austin Marckx, and Albert Montillo. scMEDAL for the interpretable analysis of single-cell transcriptomics data with batch effect visualization using a deep mixed effects autoencoder. *arXiv preprint arXiv:2411.06635*, 2024. URL <https://arxiv.org/abs/2411.06635>.
- [6] Feng ao Wang, Shaobo Chen, Yao Xuan, Junwei Liu, Qi Gao, Hongdong Zhu, Junjie Hou, Lixin Yuan, Jinyu Cheng, Chenxin Yi, Hai Wei, Yin Ma, Tao Xu, Kai Wen, and Yixue Li. Quantum-Boosted High-Fidelity Deep Learning. *arXiv preprint arXiv:2508.11190*, 2025. URL <https://arxiv.org/abs/2508.11190>.
- [7] Adrien Aumon, Myriam Lizotte, Guy Wolf, Kevin R. Moon, and Jake S. Rhodes. Forest-Guided Semantic Transport for Label-Supervised Manifold Alignment. *arXiv preprint arXiv:2602.00974*, 2026. URL <https://arxiv.org/abs/2602.00974>.
- [8] Seungheun Baek, Soyoon Park, Yan Ting Chok, Junhyun Lee, Jueon Park, Mogan Gim, and Jaewoo Kang. CRADLE-VAE: Enhancing Single-Cell Gene Perturbation Modeling with Counterfactual Reasoning-based Artifact Disentanglement. *arXiv preprint arXiv:2409.05484*, 2024. URL <https://arxiv.org/abs/2409.05484>.
- [9] Katherine Benjamin, Aneesha Bhandari, Zhouchun Shang, Yanan Xing, Yanru An, Nannan Zhang, Yong Hou, Ulrike Tillmann, Katherine R. Bull, and Heather A. Harrington. Multiscale topology classifies and quantifies cell types in subcellular spatial transcriptomics. *arXiv preprint arXiv:2212.06505*, 2022. URL <https://arxiv.org/abs/2212.06505>.
- [10] Paul Bertin, Joseph D. Viviano, Alejandro Tejada-Lapuerta, Weixu Wang, Stefan Bauer, Fabian J. Theis, and Yoshua Bengio. A scalable gene network model of regulatory dynamics in single cells. *arXiv preprint arXiv:2503.20027*, 2025. URL <https://arxiv.org/abs/2503.20027>.
- [11] Chang Bi, Kailun Bai, Xing Li, and Xuekui Zhang. HiCat: A Semi-Supervised Approach for Cell Type Annotation. *arXiv preprint arXiv:2412.06805*, 2024. URL <https://arxiv.org/abs/2412.06805>.
- [12] Francis Boabang and Samuel Asante Gyamerah. Escaping Local Optima in the Waddington Landscape: A Two-Stage TRPO-PPO Approach for Single-Cell Perturbation Analysis. *arXiv preprint arXiv:2510.13018*, 2025. URL <https://arxiv.org/abs/2510.13018>.
- [13] Joseph Boyd, Matthew Lyon, Martino Mansoldo, Christian Hurry, and Finnian Firth. Single-Cell Cross-Modal Transfer by Adversarial Fine-Tuning of Foundation Models. *arXiv preprint arXiv:2606.07676*, 2026. URL <https://arxiv.org/abs/2606.07676>.
- [14] Cassandra Burdziak, Elham Azizi, Sandhya Prabhakaran, and Dana Pe’er. A Nonparametric Multi-view Model for Estimating Cell Type-Specific Gene Regulatory Networks. *arXiv preprint arXiv:1902.08138*, 2019. URL <https://arxiv.org/abs/1902.08138>.
- [15] Keunho Byeon and Jin Tae Kwak. HEXST: Hexagonal Shifted-Window Transformer for Spatial Transcriptomics Gene Expression Prediction. *arXiv preprint arXiv:2605.04682*, 2026. URL <https://arxiv.org/abs/2605.04682>.
- [16] Xiaowen Cao, Li Xing, Elham Majd, Hua He, Junhua Gu, and Xuekui Zhang. A systematic evaluation of methods for cell phenotype classification using single-cell RNA sequencing data. *arXiv preprint arXiv:2110.00681*, 2021. URL <https://arxiv.org/abs/2110.00681>.
- [17] Alex Chen and Qing Zhou. Causal Discovery on Dependent Mixed Data with Applications to Gene Regulatory Network Inference. *arXiv preprint arXiv:2603.24783*, 2026. URL <https://arxiv.org/abs/2603.24783>.
- [18] Audrey Pei-Hsuan Chen. MoRE: Batch-Robust Multi-Omics Representations from Frozen Pre-trained Transformers. *arXiv preprint arXiv:2511.20382*, 2025. URL <https://arxiv.org/abs/2511.20382>.
- [19] Chi-Jane Chen, Yuhang Chen, Sukwon Yun, Natalie Stanley, and Tianlong Chen. Spatial Coordinates as a Cell Language: A Multi-Sentence Framework for Imaging Mass Cytometry Analysis. *arXiv preprint arXiv:2506.01918*, 2025. URL <https://arxiv.org/abs/2506.01918>.
- [20] Hegang Chen, Yuyin Lu, Yifan Zhao, Zhiming Dai, Fu Lee Wang, Qing Li, Yanghui Rao, and Yue Li. scE2TM improves single-cell embedding interpretability and reveals cellular perturbation signatures. *arXiv preprint arXiv:2507.08355*, 2025. URL <https://arxiv.org/abs/2507.08355>.
- [21] Jiawen Chen, Jianghao Zhang, Huaxiu Yao, and Yun Li. CellTypeAgent: Trustworthy cell type annotation with Large Language Models. *arXiv preprint arXiv:2505.08844*, 2025. URL <https://arxiv.org/abs/2505.08844>.
- [22] Jiayuan Chen, Thai-Hoang Pham, Yuanlong Wang, and Ping Zhang. Integrating Biological Knowledge for Robust Microscopy Image Profiling on De Novo Cell Lines. *arXiv preprint arXiv:2507.10737*, 2025. URL <https://arxiv.org/abs/2507.10737>.
- [23] Jiayuan Chen, Ruoqi Liu, Zishan Gu, and Ping Zhang. Intervention-Aware Multiscale Representation Learning from Imaging Phenomics and

- Perturbation Transcriptomics. *arXiv preprint arXiv:2604.22832*, 2026. URL <https://arxiv.org/abs/2604.22832>.
- [24] Shuizhou Chen, Lang Yu, Kedu Jin, Songming Zhang, Hao Wu, Wenxuan Huang, Sheng Xu, Quan Qian, Qin Chen, Lei Bai, Siqi Sun, and Zhangyang Gao. SCALE: Scalable Conditional Atlas-Level Endpoint transport for virtual cell perturbation prediction. *arXiv preprint arXiv:2603.17380*, 2026. URL <https://arxiv.org/abs/2603.17380>.
- [25] Xiaoqiao Chen, Sisi Chen, and Matt Thomson. Active feature selection discovers minimal gene sets for classifying cell types and disease states with single-cell mRNA-seq data. *arXiv preprint arXiv:2106.08317*, 2021. URL <https://arxiv.org/abs/2106.08317>.
- [26] Xingyu Chen, Shihao Ma, Runsheng Lin, Jiecong Lin, and Bo Wang. Ctrl-DNA: Controllable Cell-Type-Specific Regulatory DNA Design via Constrained RL. *arXiv preprint arXiv:2505.20578*, 2025. URL <https://arxiv.org/abs/2505.20578>.
- [27] Xiran Chen, Quan Zou, Qinyu Cai, Xiaofeng Chen, Weikai Li, and Yansu Wang. Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq. *arXiv preprint arXiv:2511.08996*, 2025. URL <https://arxiv.org/abs/2511.08996>.
- [28] Yanshuo Chen, Zhengmian Hu, Wei Chen, and Heng Huang. Fast and scalable Wasserstein-1 neural optimal transport solver for single-cell perturbation prediction. *arXiv preprint arXiv:2411.00614*, 2024. URL <https://arxiv.org/abs/2411.00614>.
- [29] Jiabei Cheng, Jiachen Li, Kaiyuan Yang, Hongbin Shen, and Ye Yuan. GenoHoption: Bridging Gene Network Graphs and Single-Cell Foundation Models. *arXiv preprint arXiv:2411.06331*, 2024. URL <https://arxiv.org/abs/2411.06331>.
- [30] Alexis Chevalier, Soumya Ghosh, Urvi Awasthi, James Watkins, Julia Bieniewska, Nichita Mitrea, Olga Kotova, Kirill Shkura, Andrew Noble, Michael Steinbaugh, Vijay Sadashivaiah, George Dasoulas, Julien Delile, Christoph Meier, Leonid Zhukov, Iya Khalil, Srayanta Mukherjee, and Judith Mueller. TEDDY: A Family Of Foundation Models For Understanding Single Cell Biology. *arXiv preprint arXiv:2503.03485*, 2025. URL <https://arxiv.org/abs/2503.03485>.
- [31] Changxi Chi, Yufei Huang, Jun Xia, Jiangbin Zheng, Yunfan Liu, Zelin Zang, and Stan Z. Li. Departures: Distributional Transport for Single-Cell Perturbation Prediction with Neural Schrodinger Bridges. *arXiv preprint arXiv:2511.13124*, 2025. URL <https://arxiv.org/abs/2511.13124>.
- [32] Won June Cho, Hongjun Yoon, Daeky Jeong, Hyeonjeol Lim, and Yosep Chong. \$MV_{\{Hybrid\}}\$: Improving Spatial Transcriptomics Prediction with Hybrid State Space-Vision Transformer Backbone in Pathology Vision Foundation Models. *arXiv preprint arXiv:2508.00383*, 2025. URL <https://arxiv.org/abs/2508.00383>.
- [33] Charlotte Claye, Pierre Marschall, Wassila Ouerdane, Celine Hudelot, and Julien Duquesne. Discovering Interpretable Biological Concepts in Single-cell RNA-seq Foundation Models. *arXiv preprint arXiv:2510.25807*, 2025. URL <https://arxiv.org/abs/2510.25807>.
- [34] William Connell, Umair Khan, and Michael J. Keiser. A single-cell gene expression language model. *arXiv preprint arXiv:2210.14330*, 2022. URL <https://arxiv.org/abs/2210.14330>.
- [35] Daniela Corbetta, Livio Finos, Ludwig Geistlinger, and Davide Risso. Conformal inference for cell type annotation with graph-structured constraints. *arXiv preprint arXiv:2410.23786*, 2024. URL <https://arxiv.org/abs/2410.23786>.
- [36] Omar Coser. ELISA: An Interpretable Hybrid Generative AI Agent for Expression-Grounded Discovery in Single-Cell Genomics. *arXiv preprint arXiv:2603.11872*, 2026. URL <https://arxiv.org/abs/2603.11872>.
- [37] Zhuorui Cui, Shengze Dong, and Ding Liu. White-Box Diffusion Transformer for single-cell RNA-seq generation. *arXiv preprint arXiv:2411.06785*, 2024. URL <https://arxiv.org/abs/2411.06785>.
- [38] Pedro Henrique da Costa Avelar, Min Wu, and Sophia Tsoka. Subset-Contrastive Multi-Omics Network Embedding. *arXiv preprint arXiv:2504.11321*, 2025. URL <https://arxiv.org/abs/2504.11321>.
- [39] Haoyue Dai, Ignavier Ng, Gongxu Luo, Peter Spirtes, Petar Stojanov, and Kun Zhang. Gene Regulatory Network Inference in the Presence of Dropouts: a Causal View. *arXiv preprint arXiv:2403.15500*, 2024. URL <https://arxiv.org/abs/2403.15500>.
- [40] Saleem A. Al Dajani, Abel Sanchez, and John R. Williams. DeepSeq: High-Throughput Single-Cell RNA Sequencing Data Labeling via Web Search-Augmented Agentic Generative AI Foundation Models. *arXiv preprint arXiv:2506.13817*, 2025. URL <https://arxiv.org/abs/2506.13817>.
- [41] Michael M. Danziger, Bharath Dandala, Viatcheslav Gurev, Matthew Madgwick, Sivan Ravid, Tim Rumbell, Akira Koseki, Tal Kozlovski, Ching-Huei Tsou, Ella Barkan, Tanwi Biswas, Jieli Xu, Yishai Shimoni, Jianying Hu, and Michal Rosen-Zvi. BMFM-RNA: whole-cell expression decoding improves transcriptomic foundation models. *arXiv*

- preprint *arXiv:2506.14861*, 2025. URL <https://arxiv.org/abs/2506.14861>.
- [42] Muhammed Furkan Dasdelen, Fatih Ozlugedik, Ilaria Looser, Rao Muhammad Umer, Christian Pohlkamp, and Carsten Marr. Genetically Aligned Patient Representations Improve Hematological Diagnosis. *arXiv preprint arXiv:2605.29980*, 2026. URL <https://arxiv.org/abs/2605.29980>.
 - [43] Miguel Fernandez de Retana, Ruben Sanchez-Corcuera, Unai Zulaika, Aritz Bilbao-Jayo, and Aitor Almeida. When Does Gene Regulatory Network Inference Break? A Controlled Diagnostic Study of Causal and Correlational Methods on Single-Cell Data. *arXiv preprint arXiv:2605.04930*, 2026. URL <https://arxiv.org/abs/2605.04930>.
 - [44] Andac Demir, Elizaveta Solovyeva, James Boylan, Mei Xiao, Fabrizio Serluca, Sebastian Hoersch, Jeremy Jenkins, Murthy Devarakonda, and Bulent Kiziltan. sc-OTGM: Single-Cell Perturbation Modeling by Solving Optimal Mass Transport on the Manifold of Gaussian Mixtures. *arXiv preprint arXiv:2405.03726*, 2024. URL <https://arxiv.org/abs/2405.03726>.
 - [45] Andac Demir, Erik W. Anderson, Jeremy L. Jenkins, and Srayanta Mukherjee. CellxPert: Inference-Time MCMC Steering of a Multi-Omics Single-Cell Foundation Model for In-Silico Perturbation. *arXiv preprint arXiv:2605.00930*, 2026. URL <https://arxiv.org/abs/2605.00930>.
 - [46] Manuel Dileo and Andrea Sottoriva. Applications of temporal graph learning for predicting the dynamics of biological systems. *arXiv preprint arXiv:2605.28659*, 2026. URL <https://arxiv.org/abs/2605.28659>.
 - [47] Sajib Acharjee Dip and Liqing Zhang. Stable-Shift: Biologically Structured Prediction of Transcriptional Responses to Unseen Gene Perturbations. *arXiv preprint arXiv:2606.24940*, 2026. URL <https://arxiv.org/abs/2606.24940>.
 - [48] Sajib Acharjee Dip, Adrika Zafor, Bikash Kumar Paul, Uddip Acharjee Shuvo, Muhit Islam Emon, Xuan Wang, and Liqing Zhang. LLM4Cell: A Survey of Large Language and Agentic Models for Single-Cell Biology. *arXiv preprint arXiv:2510.07793*, 2025. URL <https://arxiv.org/abs/2510.07793>.
 - [49] Shengze Dong, Zhuorui Cui, Ding Liu, and Jinzhi Lei. scRDiT: Generating single-cell RNA-seq data by diffusion transformers and accelerating sampling. *arXiv preprint arXiv:2404.06153*, 2024. URL <https://arxiv.org/abs/2404.06153>.
 - [50] Xishuang Dong, Shanta Chowdhury, Uboho Victor, Xiangfang Li, and Lijun Qian. Cell Type Identification from Single-Cell Transcriptomic Data via Semi-supervised Learning. *arXiv preprint arXiv:2005.03994*, 2020. URL <https://arxiv.org/abs/2005.03994>.
 - [51] Ziyang Dong, Shanwen Tan, Hengchuang Yin, Wei Liu, Yifan Wang, Siyu Yi, Jiancheng Lv, and Wei Ju. BRIDGE: Biological Evidence Refinement and Heterogeneous Dynamic Gating for Gene Regulatory Networks. *arXiv preprint arXiv:2606.14734*, 2026. URL <https://arxiv.org/abs/2606.14734>.
 - [52] Yi Duan, Zhao Yang, Jiwei Zhu, Ying Ba, Chuan Cao, and Bing Su. Biological Reasoning-Informed Regression for Interpretable Regulatory DNA Activity Prediction. *arXiv preprint arXiv:2606.08147*, 2026. URL <https://arxiv.org/abs/2606.08147>.
 - [53] Ali ElSheikh, Rui-Xi Wang, Weimin Wu, Yibo Wen, Payam Dibaeinia, Jennifer Yuntong Zhang, Jerry Yao-Chieh Hu, Mei Knudson, Sudarshan Babu, Shao-Hua Sun, Aly A. Khan, and Han Liu. Cell-JEPA: Latent Representation Learning for Single-Cell Transcriptomics. *arXiv preprint arXiv:2602.02093*, 2026. URL <https://arxiv.org/abs/2602.02093>.
 - [54] Jan Engelmann, Leon Hetzel, Giovanni Palla, Lisa Sikkema, Malte Luecken, and Fabian Theis. Uncertainty Quantification for Atlas-Level Cell Type Transfer. *arXiv preprint arXiv:2211.03793*, 2022. URL <https://arxiv.org/abs/2211.03793>.
 - [55] Congyu Fang, Adam Dziedzic, Lin Zhang, Laura Oliva, Amol Verma, Fahad Razak, Nicolas Papernot, and Bo Wang. Decentralised, Collaborative, and Privacy-preserving Machine Learning for Multi-Hospital Data. *arXiv preprint arXiv:2402.00205*, 2024. URL <https://arxiv.org/abs/2402.00205>.
 - [56] Donghai Fang, Yongheng Li, Zhen Wang, Yuan-song Zeng, and Wenwen Min. Adapting a Pre-trained Single-Cell Foundation Model to Spatial Gene Expression Generation from Histology Images. *arXiv preprint arXiv:2603.19766*, 2026. URL <https://arxiv.org/abs/2603.19766>.
 - [57] Yin Fang, Xinle Deng, Kangwei Liu, Ningyu Zhang, Jingyang Qian, Penghui Yang, Xiaohui Fan, and Huajun Chen. A Multi-Modal AI Copilot for Single-Cell Analysis with Instruction Following. *arXiv preprint arXiv:2501.08187*, 2025. URL <https://arxiv.org/abs/2501.08187>.
 - [58] Yin Fang, Qiao Jin, Guangzhi Xiong, Bowen Jin, Xianrui Zhong, Siru Ouyang, Aidong Zhang, Jiawei Han, and Zhiyong Lu. Cell-ol: Training LLMs to Solve Single-Cell Reasoning Puzzles with Reinforcement Learning. *arXiv preprint arXiv:2506.02911*, 2025. URL <https://arxiv.org/abs/2506.02911>.
 - [59] Justin Feigelman, Fabian J. Theis, and Carsten Marr. MCA: Multiresolution Correlation Analysis, a graphical tool for subpopulation identification in

- single-cell gene expression data. *arXiv preprint arXiv:1407.2112*, 2014. URL <https://arxiv.org/abs/1407.2112>.
- [60] Jasmin Fisher, Ali Sinan Koksai, Nir Piterman, and Steven Woodhouse. Synthesising Executable Gene Regulatory Networks from Single-cell Gene Expression Data. *arXiv preprint arXiv:1505.05193*, 2015. URL <https://arxiv.org/abs/1505.05193>.
- [61] Adam Foster, Arpi Vezer, Craig A Glastonbury, Paidi Creed, Sam Abujudeh, and Aaron Sim. Contrastive Mixture of Posteriors for Counterfactual Inference, Data Integration and Fairness. *arXiv preprint arXiv:2106.08161*, 2021. URL <https://arxiv.org/abs/2106.08161>.
- [62] Andrea Giuseppe Di Francesco, Andrea Rubbi, and Pietro Lio. Retrieval-Augmented Generation for Predicting Cellular Responses to Gene Perturbation. *arXiv preprint arXiv:2603.07233*, 2026. URL <https://arxiv.org/abs/2603.07233>.
- [63] Boyang Fu, George Dasoulas, Sameer Gabbita, Xiang Lin, Shanghua Gao, Xiaorui Su, Soumya Ghosh, and Marinka Zitnik. STRAND: Sequence-Conditioned Transport for Single-Cell Perturbations. *arXiv preprint arXiv:2602.10156*, 2026. URL <https://arxiv.org/abs/2602.10156>.
- [64] Yiming Gao, Zhen Wang, Jefferson Chen, Mark Antkowiak, Mengzhou Hu, JungHo Kong, Dexter Pratt, Jieyuan Liu, Enze Ma, Zhiting Hu, and Eric P. Xing. scPilot: Large Language Model Reasoning Toward Automated Single-Cell Analysis and Discovery. *arXiv preprint arXiv:2602.11609*, 2026. URL <https://arxiv.org/abs/2602.11609>.
- [65] William Gilpin. The cell as a token: high-dimensional geometry in language models and cell embeddings. *arXiv preprint arXiv:2503.20278*, 2025. URL <https://arxiv.org/abs/2503.20278>.
- [66] Jing Gong, Minsheng Hao, Xingyi Cheng, Xin Zeng, Chiming Liu, Jianzhu Ma, Xuegong Zhang, Taifeng Wang, and Le Song. xTrimGene: An Efficient and Scalable Representation Learner for Single-Cell RNA-Seq Data. *arXiv preprint arXiv:2311.15156*, 2023. URL <https://arxiv.org/abs/2311.15156>.
- [67] Gokul Gowri, Igor Sadalski, Dan Raviv, Peng Yin, Jonathan Rosenfeld, and Allon M. Klein. A measurement noise scaling law for cellular representation learning. *arXiv preprint arXiv:2503.02726*, 2025. URL <https://arxiv.org/abs/2503.02726>.
- [68] Ehsan Hajiramezanali, Mahdi Imani, Ulisses Braga-Neto, Xiaoning Qian, and Edward R Dougherty. Scalable optimal Bayesian classification of single-cell trajectories under regulatory model uncertainty. *arXiv preprint arXiv:1902.03188*, 2019. URL <https://arxiv.org/abs/1902.03188>.
- [69] Dezheng Han, Yibin Jia, Ruxiao Chen, Wenjie Han, Shuaishuai Guo, and Jianbo Wang. ReCellTy: Domain-Specific Knowledge Graph Retrieval-Augmented LLMs Reasoning Workflow for Single-Cell Annotation. *arXiv preprint arXiv:2505.00017*, 2025. URL <https://arxiv.org/abs/2505.00017>.
- [70] Jiashu Han, Kunzan Liu, Yeojin Kim, Saurabh Sinha, and Sixian You. MAD: Microenvironment-Aware Distillation – A Pretraining Strategy for Virtual Spatial Omics from Microscopy. *arXiv preprint arXiv:2603.13401*, 2026. URL <https://arxiv.org/abs/2603.13401>.
- [71] Minghao Han, Dingkan Yang, Jiabei Cheng, Xukun Zhang, Linhao Qu, Zizhi Chen, and Lihua Zhang. Towards Unified Molecule-Enhanced Pathology Image Representation Learning via Integrating Spatial Transcriptomics. *arXiv preprint arXiv:2412.00651*, 2024. URL <https://arxiv.org/abs/2412.00651>.
- [72] Seong Woo Han, Daniel Duy Vo, and Brielin C. Brown. Large-Scale Bayesian Causal Discovery with Interventional Data. *arXiv preprint arXiv:2510.01562*, 2025. URL <https://arxiv.org/abs/2510.01562>.
- [73] Ninghui Hao, Xinxing Yang, Boshen Yan, Dong Li, Junzhou Huang, Xintao Wu, Emily S. Ruiz, Arlene Ruiz de Luzuriaga, Chen Zhao, and Guihong Wan. Histopathology-centered Computational Evolution of Spatial Omics: Integration, Mapping, and Foundation Models. *arXiv preprint arXiv:2601.07826*, 2026. URL <https://arxiv.org/abs/2601.07826>.
- [74] Euxhen Hasanaj, Barnabas Poczos, and Ziv Bar-Joseph. Recovering Time-Varying Networks From Single-Cell Data. *arXiv preprint arXiv:2410.01853*, 2024. URL <https://arxiv.org/abs/2410.01853>.
- [75] Suvojit Hazra, Marina Kotvanova, Kirstan Gimse, and Sushmita Roy. MERLIN-SUITE: Probabilistic modular GRN inference from multi-omics data integrating regulatory priors and transcription factor activity. *arXiv preprint arXiv:2607.01791*, 2026. URL <https://arxiv.org/abs/2607.01791>.
- [76] Suvojit Hazra, Chandrani Kumari, and Sushmita Roy. scMTNI: Leveraging cellular trajectory and context to infer dynamic GRNs from single-cell multi-omics data. *arXiv preprint arXiv:2607.01508*, 2026. URL <https://arxiv.org/abs/2607.01508>.
- [77] Bing He, Yao Xiao, Haodong Liang, Qianhui Huang, Yuheng Du, Yijun Li, David Garmire, Duxin Sun, and Lana X. Garmire. ASGARD: A Single-cell Guided pipeline to Aid Repurposing of Drugs. *arXiv preprint arXiv:2109.06377*, 2021. URL <https://arxiv.org/abs/2109.06377>.

- [78] Yongjian He, Vered Klein, Orr Levy, and Xu-Wen Wang. Predicting cell-specific gene expression profile and knockout impact through deep learning. *arXiv preprint arXiv:2510.03359*, 2025. URL <https://arxiv.org/abs/2510.03359>.
- [79] Konstantin Hemker, Andrew H. Song, Cristina Almagro-Perez, Guillaume Jaume, Sophia J. Wagner, Anurag Vaidya, Nikola Simidjievski, Mateja Jamnik, and Faisal Mahmood. Towards Spatial Transcriptomics-driven Pathology Foundation Models. *arXiv preprint arXiv:2602.14177*, 2026. URL <https://arxiv.org/abs/2602.14177>.
- [80] Ulysse Herbach. Gene regulatory network inference from single-cell data using a self-consistent proteomic field. *arXiv preprint arXiv:2109.14888*, 2021. URL <https://arxiv.org/abs/2109.14888>.
- [81] Ulysse Herbach. Harissa: stochastic simulation and inference of gene regulatory networks based on transcriptional bursting. *arXiv preprint arXiv:2309.05112*, 2023. URL <https://arxiv.org/abs/2309.05112>.
- [82] Fabian Horst, Moritz Rempe, Helmut Becker, Lukas Heine, Julius Keyl, and Jens Kleesiek. CellViT++: Energy-Efficient and Adaptive Cell Segmentation and Classification Using Foundation Models. *arXiv preprint arXiv:2501.05269*, 2025. URL <https://arxiv.org/abs/2501.05269>.
- [83] Sosuke Hosokawa, Toshiharu Kawakami, Satoshi Kadera, Masamichi Ito, and Norihiko Takeda. Transcoder-based Circuit Analysis for Interpretable Single-Cell Foundation Models. *arXiv preprint arXiv:2509.14723*, 2025. URL <https://arxiv.org/abs/2509.14723>.
- [84] Susu Hu and Stefanie Speidel. MoLF: Mixture-of-Latent-Flow for Pan-Cancer Spatial Gene Expression Prediction from Histology. *arXiv preprint arXiv:2602.02282*, 2026. URL <https://arxiv.org/abs/2602.02282>.
- [85] Tianxiang Hu, Chenyi Zhou, Jiaxiang Liu, Jiongxin Wang, Ruizhe Chen, Haoxiang Xia, Gaoang Wang, Jian Wu, and Zuozhu Liu. GRIT: Graph-Regularized Logit Refinement for Zero-shot Cell Type Annotation. *arXiv preprint arXiv:2508.04747*, 2025. URL <https://arxiv.org/abs/2508.04747>.
- [86] Chao Hui Huang. QuST-LLM: Integrating Large Language Models for Comprehensive Spatial Transcriptomics Analysis. *arXiv preprint arXiv:2406.14307*, 2024. URL <https://arxiv.org/abs/2406.14307>.
- [87] Min Huang and Rishikesan Kamaleswaran. SpikGPT: A High-Accuracy and Interpretable Spiking Attention Framework for Single-Cell Annotation. *arXiv preprint arXiv:2512.03286*, 2025. URL <https://arxiv.org/abs/2512.03286>.
- [88] Tinglin Huang, Tianyu Liu, Mehrtash Babadi, Wengong Jin, and Rex Ying. Scalable Generation of Spatial Transcriptomics from Histology Images via Whole-Slide Flow Matching. *arXiv preprint arXiv:2506.05361*, 2025. URL <https://arxiv.org/abs/2506.05361>.
- [89] Xiaoshui Huang, Tianlin Zhu, Yifan Zuo, Xue Xia, Zonghan Wu, Jiebin Yan, Dingli Hua, Zongyi Xu, Yuming Fang, and Jian Zhang. PanFoMa: A Lightweight Foundation Model and Benchmark for Pan-Cancer. *arXiv preprint arXiv:2512.03111*, 2025. URL <https://arxiv.org/abs/2512.03111>.
- [90] Yu-An Huang, Xiyue Cao, Zhu-Hong You, Yue-Chao Li, Xuequn Shang, and Zhi-An Huang. scGSDR: Harnessing Gene Semantics for Single-Cell Pharmacological Profiling. *arXiv preprint arXiv:2502.01689*, 2025. URL <https://arxiv.org/abs/2502.01689>.
- [91] Yu-An Huang, Yue-Chao Li, Hai-Ru You, Jie Pan, Xiyue Cao, Xinyuan Li, Zhi-An Huang, and Zhu-Hong You. Graph Structure Learning for Tumor Microenvironment with Cell Type Annotation from non-spatial scRNA-seq data. *arXiv preprint arXiv:2502.02629*, 2025. URL <https://arxiv.org/abs/2502.02629>.
- [92] Yuichiro Iwashita, Ahtisham Fazeel Abbasi, Koichi Kise, Andreas Dengel, and Muhammad Nabeel Asim. A Large-Scale Comparative Analysis of Imputation Methods for Single-Cell RNA Sequencing Data. *arXiv preprint arXiv:2603.24626*, 2026. URL <https://arxiv.org/abs/2603.24626>.
- [93] Danning Jiang, Zheming An, Yalong Zhao, and Lipeng Lai. OCOO-T : A Simple and Scalable Virtual Cell Model for Transcriptional Perturbation Response Prediction. *arXiv preprint arXiv:2606.12838*, 2026. URL <https://arxiv.org/abs/2606.12838>.
- [94] Douglas Jiang, Zilin Dai, Luxuan Zhang, Qiyi Yu, Haoqi Sun, and Feng Tian. Bridging Large Language Models and Single-Cell Transcriptomics in Dissecting Selective Motor Neuron Vulnerability. *arXiv preprint arXiv:2505.07896*, 2025. URL <https://arxiv.org/abs/2505.07896>.
- [95] Jialong Jiang, Sisi Chen, Tiffany Tsou, Christopher S. McGinnis, Tahmineh Khazaei, Qin Zhu, Jong H. Park, Paul Rivaud, Inna-Marie Strazhnik, Eric D. Chow, David A. Sivak, Zev J. Gartner, and Matt Thomson. Therapeutic algebra of immunomodulatory drug responses at single-cell resolution. *arXiv preprint arXiv:2208.10661*, 2022. URL <https://arxiv.org/abs/2208.10661>.
- [96] Qi Jiang, Lei Zhang, Longquan Li, and Lin Wan. Learning collective multi-cellular dynamics from temporal scRNA-seq via a transformer-enhanced

- Neural SDE. *arXiv preprint arXiv:2505.16492*, 2025. URL <https://arxiv.org/abs/2505.16492>.
- [97] Weikang Jiang, Jinxian Wang, Jihong Guan, and Shuigeng Zhou. scRNA-seq Data Clustering by Cluster-aware Iterative Contrastive Learning. *arXiv preprint arXiv:2312.16600*, 2023. URL <https://arxiv.org/abs/2312.16600>.
- [98] Wenkang Jiang, Yuhang Liu, Yichao Cai, Erdun Gao, Jiayi Dong, Ehsan Abbasnejad, Lina Yao, and Javen Qinfeng Shi. What Makes a Representation Good for Single-Cell Perturbation Prediction? *arXiv preprint arXiv:2605.19343*, 2026. URL <https://arxiv.org/abs/2605.19343>.
- [99] Wenkang Jiang, Yuhang Liu, Erdun Gao, Ehsan Abbasnejad, Lina Yao, and Javen Qinfeng Shi. Learning Latent Dynamical Causal Processes for Single-Cell Perturbation Prediction. *arXiv preprint arXiv:2605.25581*, 2026. URL <https://arxiv.org/abs/2605.25581>.
- [100] Yifeng Jiao, Yuchen Liu, Yu Zhang, Xin Guo, Yushuai Wu, Chen Jiang, Jiyang Li, Hongwei Zhang, Limei Han, Xin Gao, Yuan Qi, and Yuan Cheng. ChromFound: Towards A Universal Foundation Model for Single-Cell Chromatin Accessibility Data. *arXiv preprint arXiv:2505.12638*, 2025. URL <https://arxiv.org/abs/2505.12638>.
- [101] Ihor Kendiukhov. Causal Circuit Tracing Reveals Distinct Computational Architectures in Single-Cell Foundation Models: Inhibitory Dominance, Biological Coherence, and Cross-Model Convergence. *arXiv preprint arXiv:2603.01752*, 2026. URL <https://arxiv.org/abs/2603.01752>.
- [102] Ihor Kendiukhov. Multi-Dimensional Spectral Geometry of Biological Knowledge in Single-Cell Transformer Representations. *arXiv preprint arXiv:2602.22247*, 2026. URL <https://arxiv.org/abs/2602.22247>.
- [103] Ihor Kendiukhov. Quantifying Ranking Instability Across Evaluation Protocol Axes in Gene Regulatory Network Benchmarking. *arXiv preprint arXiv:2603.03493*, 2026. URL <https://arxiv.org/abs/2603.03493>.
- [104] Ihor Kendiukhov. Scaling Laws for Masked-Reconstruction Transformers on Single-Cell Transcriptomics. *arXiv preprint arXiv:2602.15253*, 2026. URL <https://arxiv.org/abs/2602.15253>.
- [105] Ihor Kendiukhov. Sparse autoencoders reveal organized biological knowledge but minimal regulatory logic in single-cell foundation models: a comparative atlas of Geneformer and scGPT. *arXiv preprint arXiv:2603.02952*, 2026. URL <https://arxiv.org/abs/2603.02952>.
- [106] Ihor Kendiukhov. Systematic Evaluation of Single-Cell Foundation Model Interpretability Reveals Attention Captures Co-Expression Rather Than Unique Regulatory Signal. *arXiv preprint arXiv:2602.17532*, 2026. URL <https://arxiv.org/abs/2602.17532>.
- [107] Ihor Kendiukhov. What Topological and Geometric Structure Do Biological Foundation Models Learn? Evidence from 141 Hypotheses. *arXiv preprint arXiv:2602.22289*, 2026. URL <https://arxiv.org/abs/2602.22289>.
- [108] Oussama Kharouiche, Aris Markogiannakis, Xiao Fei, Michail Chatzianastasis, and Michalis Vazirgiannis. Cell2Text: Multimodal LLM for Generating Single-Cell Descriptions from RNA-Seq Data. *arXiv preprint arXiv:2509.24840*, 2025. URL <https://arxiv.org/abs/2509.24840>.
- [109] Hyomin Kim, Sang-Yeon Hwang, Jaechang Lim, Yinhua Piao, Yunhak Oh, Woo Youn Kim, Chanyoung Park, Sungsoo Ahn, and Junhyeok Jeon. Progressive Multi-Agent Reasoning for Biological Perturbation Prediction. *arXiv preprint arXiv:2602.07408*, 2026. URL <https://arxiv.org/abs/2602.07408>.
- [110] Sindhura Kommu, Yizhi Wang, Yue Wang, and Xuan Wang. Gene Regulatory Network Inference from Pre-trained Single-Cell Transcriptomics Transformer with Joint Graph Learning. *arXiv preprint arXiv:2407.18181*, 2024. URL <https://arxiv.org/abs/2407.18181>.
- [111] Da Kuang, Guanwen Qiu, and Junhyong Kim. Reconstructing Cell Lineage Trees from Phenotypic Features with Metric Learning. *arXiv preprint arXiv:2503.13925*, 2025. URL <https://arxiv.org/abs/2503.13925>.
- [112] Wei Lan, Guohang He, Mingyang Liu, Qingfeng Chen, Junyue Cao, and Wei Peng. Transformer-based Single-Cell Language Model: A Survey. *arXiv preprint arXiv:2407.13205*, 2024. URL <https://arxiv.org/abs/2407.13205>.
- [113] Olatz Sanz Larrarte, Borja Aizpurua, Reza Dastbaste, Ruben M. Otxoa, and Josu Etzezarreta Martinez. Tensor Network based Gene Regulatory Network Inference for Single-Cell Transcriptomic Data. *arXiv preprint arXiv:2509.06891*, 2025. URL <https://arxiv.org/abs/2509.06891>.
- [114] Elliot Layne, Jason Hartford, Sebastien Lachapelle, Mathieu Blanchette, and Dhanya Sridhar. Sparsity regularization via tree-structured environments for disentangled representations. *arXiv preprint arXiv:2405.20482*, 2024. URL <https://arxiv.org/abs/2405.20482>.
- [115] Konstantinos Lazaros, Dimitris E. Koumadorakis, Panagiotis Vlamos, and Aristidis G. Vrahatis. Graph Neural Network approaches for single-cell data: A recent overview. *arXiv preprint*

- arXiv:2310.09561*, 2023. URL <https://arxiv.org/abs/2310.09561>.
- [116] Minh H. N. Le, Tuan Vinh, Thanh-Huy Nguyen, Tao Li, Bao Quang Gia Le, Han H. Huynh, Monika Raj, Carl Yang, Min Xu, and Nguyen Quoc Khanh Le. Imaging-anchored Multiomics in Cardiovascular Disease: Integrating Cardiac Imaging, Bulk, Single-cell, and Spatial Transcriptomics. *arXiv preprint arXiv:2601.07871*, 2026. URL <https://arxiv.org/abs/2601.07871>.
 - [117] Minsoo Lee, Jonghyun Kim, Juseung Yun, Sunwoo Yu, and Jongseong Jang. MINT: Molecularly Informed Training with Spatial Transcriptomics Supervision for Pathology Foundation Models. *arXiv preprint arXiv:2603.07895*, 2026. URL <https://arxiv.org/abs/2603.07895>.
 - [118] Cong Li, Qingqing Long, Yuanchun Zhou, and Meng Xiao. scReader: Prompting Large Language Models to Interpret scRNA-seq Data. *arXiv preprint arXiv:2412.18156*, 2024. URL <https://arxiv.org/abs/2412.18156>.
 - [119] Cong Li, Meng Xiao, Pengfei Wang, Guihai Feng, Xin Li, and Yuanchun Zhou. scInterpreter: Training Large Language Models to Interpret scRNA-seq Data for Cell Type Annotation. *arXiv preprint arXiv:2402.12405*, 2024. URL <https://arxiv.org/abs/2402.12405>.
 - [120] Huifa Li, Jie Fu, Xinpeng Ling, Zhiyu Sun, Kuncan Wang, and Zhili Chen. Single-cell Curriculum Learning-based Deep Graph Embedding Clustering. *arXiv preprint arXiv:2408.10511*, 2024. URL <https://arxiv.org/abs/2408.10511>.
 - [121] Huifa Li, Jie Fu, Xinlin Zhuang, Haolin Yang, Xinpeng Ling, Tong Cheng, Haochen xue, Imran Razak, and Zhili Chen. scAGC: Learning Adaptive Cell Graphs with Contrastive Guidance for Single-Cell Clustering. *arXiv preprint arXiv:2508.09180*, 2025. URL <https://arxiv.org/abs/2508.09180>.
 - [122] Jiahao Li, Jiayi Dong, Peng Ye, Xiaochi Zhou, Hao-hai Lu, and Fei Wang. SAVE: A Generalizable Framework for Multi-Condition Single-Cell Generation with Gene Block Attention. *arXiv preprint arXiv:2604.16776*, 2026. URL <https://arxiv.org/abs/2604.16776>.
 - [123] Jiaxin Li and Shanjun Mao. Time-Varying Network Driver Estimation (TNDE) Quantifies Stage-Specific Regulatory Effects From Single-Cell Snapshots. *arXiv preprint arXiv:2511.19813*, 2025. URL <https://arxiv.org/abs/2511.19813>.
 - [124] Rongbin Li, Wenbo Chen, Zhao Li, Rodrigo Munoz-Castaneda, Jinbo Li, Neha S. Maurya, Arnav Solanki, Huan He, Hanwen Xing, Meaghan Ramlakhan, Zachary Wise, Nelson Johansen, Zhuohao Wu, Hua Xu, Michael Hawrylycz, and W. Jim Zheng. BRAINCELL-AID: An Agentic AI Created Brain Cell Type Resource for Community Annotation. *arXiv preprint arXiv:2510.17064*, 2025. URL <https://arxiv.org/abs/2510.17064>.
 - [125] Xiaoyu Li, Fangfang Zhu, and Wenwen Min. SpaDiT: Diffusion Transformer for Spatial Gene Expression Prediction using scRNA-seq. *arXiv preprint arXiv:2407.13182*, 2024. URL <https://arxiv.org/abs/2407.13182>.
 - [126] Yixuan Li, Archer Y. Yang, and Yue Li. Disentangling Shared and Target-Enriched Topics via Background-Contrastive Non-negative Matrix Factorization. *arXiv preprint arXiv:2602.22387*, 2026. URL <https://arxiv.org/abs/2602.22387>.
 - [127] Zhaokang Liang, Shuyang Zhuang, Xiaoran Jiao, Weian Mao, Hao Chen, and Chunhua Shen. scPPDM: A Diffusion Model for Single-Cell Drug-Response Prediction. *arXiv preprint arXiv:2510.11726*, 2025. URL <https://arxiv.org/abs/2510.11726>.
 - [128] Ling Liao, Changhui Yang, Maxim Artyomov, Mark Watson, Adam Kepecs, Haowen Zhou, Alexey Sergushichev, and Richard Cote. Digital Modeling of Spatial Pathway Activity from Histology Reveals Tumor Microenvironment Heterogeneity. *arXiv preprint arXiv:2512.09003*, 2025. URL <https://arxiv.org/abs/2512.09003>.
 - [129] Hui Liu, Wei Duan, and Judong Luo. Towards generalization of drug response prediction to single cells and patients utilizing importance-aware multi-source domain transfer learning. *arXiv preprint arXiv:2403.05260*, 2024. URL <https://arxiv.org/abs/2403.05260>.
 - [130] Jiajia Liu, Mengyuan Yang, Yankai Yu, Haixia Xu, Tianguang Wang, Kang Li, and Xiaobo Zhou. Advancing bioinformatics with large language models: components, applications and perspectives. *arXiv preprint arXiv:2401.04155*, 2024. URL <https://arxiv.org/abs/2401.04155>.
 - [131] Junhao Liu, Siwei Xu, Lei Zhang, and Jing Zhang. Single-Cell Omics Arena: A Benchmark Study for Large Language Models on Cell Type Annotation Using Single-Cell Data. *arXiv preprint arXiv:2412.02915*, 2024. URL <https://arxiv.org/abs/2412.02915>.
 - [132] Xiaoxia Liu, Robert R Butler, Prashna K Gyawali, Frank M Longo, and Zihuai He. ScAtt: an Attention based architecture to analyze Alzheimer’s disease at cell type level from single-cell RNA-sequencing data. *arXiv preprint arXiv:2405.17433*, 2024. URL <https://arxiv.org/abs/2405.17433>.
 - [133] Jiangkai Long, Yanran Zhu, Chang Tang, Kun Sun, Yuanyuan Liu, and Xuesong Yan. When Genes Speak: A Semantic-Guided Framework for

- Spatially Resolved Transcriptomics Data Clustering. *arXiv preprint arXiv:2511.11380*, 2025. URL <https://arxiv.org/abs/2511.11380>.
- [134] Romain Lopez, Natasa Tagasovska, Stephen Ra, Kyunghyn Cho, Jonathan K. Pritchard, and Aviv Regev. Learning Causal Representations of Single Cells via Sparse Mechanism Shift Modeling. *arXiv preprint arXiv:2211.03553*, 2022. URL <https://arxiv.org/abs/2211.03553>.
- [135] Mohammad Lotfollahi, Mohsen Naghipourfar, Fabian J. Theis, and F. Alexander Wolf. Conditional out-of-sample generation for unpaired data using trVAE. *arXiv preprint arXiv:1910.01791*, 2019. URL <https://arxiv.org/abs/1910.01791>.
- [136] Erpai Luo, Minsheng Hao, Lei Wei, and Xuegong Zhang. scDiffusion: conditional generation of high-quality single-cell data using diffusion model. *arXiv preprint arXiv:2401.03968*, 2024. URL <https://arxiv.org/abs/2401.03968>.
- [137] Erpai Luo, Jinmeng Jia, Yifan Xiong, Xiangyu Li, Xiaobo Guo, Baoqi Yu, Minsheng Hao, Lei Wei, and Xuegong Zhang. Benchmarking AI scientists for omics data driven biological discovery. *arXiv preprint arXiv:2505.08341*, 2025. URL <https://arxiv.org/abs/2505.08341>.
- [138] Suryanarayana Maddu, Victor Chardes, and Michael J. Shelley. Learning biophysical models of gene regulation with probability flow matching. *arXiv preprint arXiv:2604.25062*, 2026. URL <https://arxiv.org/abs/2604.25062>.
- [139] Hiren Madhu, Joao Felipe Rocha, Tinglin Huang, Siddharth Viswanath, Smita Krishnaswamy, and Rex Ying. HEIST: A Graph Foundation Model for Spatial Transcriptomics and Proteomics Data. *arXiv preprint arXiv:2506.11152*, 2025. URL <https://arxiv.org/abs/2506.11152>.
- [140] Sepideh Maleki, Jan-Christian Huetter, Kangway V. Chuang, David Richmond, Gabriele Scalia, and Tommaso Biancalani. Efficient Fine-Tuning of Single-Cell Foundation Models Enables Zero-Shot Molecular Perturbation Prediction. *arXiv preprint arXiv:2412.13478*, 2024. URL <https://arxiv.org/abs/2412.13478>.
- [141] Yuren Mao, Yu Mi, Peigen Liu, Mengfei Zhang, Hanqing Liu, and Yunjun Gao. scAgent: Universal Single-Cell Annotation via a LLM Agent. *arXiv preprint arXiv:2504.04698*, 2025. URL <https://arxiv.org/abs/2504.04698>.
- [142] Marc Bounnovski Martell, Josefa Lia Stoisser, Lawrence Phillips, Aditya Misra, Robert Kitchen, Jesper Ferkinghoff-Borg, Jialin Yu, Philip Torr, and Kaspar Marten. MechPert: Mechanistic Consensus as an Inductive Bias for Unseen Perturbation Prediction. *arXiv preprint arXiv:2602.13791*, 2026. URL <https://arxiv.org/abs/2602.13791>.
- [143] Stathis Megas, Daniel G. Chen, Krzysztof Polanski, Moshe Eliasof, Carola-Bibiane Schonlieb, and Sarah A. Teichmann. Celcomen: spatial causal disentanglement for single-cell and tissue perturbation modeling. *arXiv preprint arXiv:2409.05804*, 2024. URL <https://arxiv.org/abs/2409.05804>.
- [144] Gabriel Mejia, Daniela Ruiz, Paula Cardenas, Leonardo Manrique, Daniela Vega, and Pablo Arbelaez. SpaRED benchmark: Enhancing Gene Expression Prediction from Histology Images with Spatial Transcriptomics Completion. *arXiv preprint arXiv:2407.13027*, 2024. URL <https://arxiv.org/abs/2407.13027>.
- [145] Zhichao Miao, Pablo Moreno, Ni Huang, Irene Papatheodorou, Alvis Brazma, and Sarah A Teichmann. Putative cell type discovery from single-cell gene expression data. *arXiv preprint arXiv:2004.09847*, 2020. URL <https://arxiv.org/abs/2004.09847>.
- [146] Mikele Milia, Louis Fabrice Tshimanga, Henning Mueller, Manfredo Atzori, and Barbara Di Camillo. Integrating gene regulatory priors into Transformer attention with scTransformer for interpretable scRNA-seq analysis. *arXiv preprint arXiv:2606.09558*, 2026. URL <https://arxiv.org/abs/2606.09558>.
- [147] Wenwen Min, Zhiceng Shi, Jun Zhang, Jun Wan, and Changmiao Wang. Multimodal contrastive learning for spatial gene expression prediction using histology images. *arXiv preprint arXiv:2407.08216*, 2024. URL <https://arxiv.org/abs/2407.08216>.
- [148] Wenwen Min, Zhen Wang, Fangfang Zhu, Taosheng Xu, and Shunfang Wang. scASDC: Attention Enhanced Structural Deep Clustering for Single-cell RNA-seq Data. *arXiv preprint arXiv:2408.05258*, 2024. URL <https://arxiv.org/abs/2408.05258>.
- [149] Maria Mircea, Diego Garlaschelli, and Stefan Semrau. Inference of dynamical gene regulatory networks from single-cell data with physics informed neural networks. *arXiv preprint arXiv:2401.07379*, 2024. URL <https://arxiv.org/abs/2401.07379>.
- [150] Dylan Molho, Jiayuan Ding, Zhaoheng Li, Hongzhi Wen, Wenzhuo Tang, Yixin Wang, Julian Venegas, Wei Jin, Renming Liu, Runze Su, Patrick Danaher, Robert Yang, Yu Leo Lei, Yuying Xie, and Jiliang Tang. Deep Learning in Single-Cell Analysis. *arXiv preprint arXiv:2210.12385*, 2022. URL <https://arxiv.org/abs/2210.12385>.
- [151] Jonathan Muhire. Donor-Aware scRNA-seq Benchmarks for IBD Classification. *arXiv preprint arXiv:2605.03281*, 2026. URL <https://arxiv.org/abs/2605.03281>.

- [152] Dinh-Nguyen Nguyen, Sadia Shakil, Raymond Kai-Yu Tong, and Ngoc-Duy Dinh. Large Language Model Agents Enable Autonomous Design and Image Analysis of Microwell Microfluidics. *arXiv preprint arXiv:2510.13883*, 2025. URL <https://arxiv.org/abs/2510.13883>.
- [153] Quang-Huy Nguyen, Jiaqi Wang, and Wei-Shinn Ku. scBatchProx: Federated-Inspired Refinement for Stable Cell-Type Discriminability under Heterogeneous Batch Compositions. *arXiv preprint arXiv:2602.00423*, 2026. URL <https://arxiv.org/abs/2602.00423>.
- [154] Yuki Nishio, Takayoshi Yamashita, Keita Ito, Tsubasa Hirakawa, and Hironobu Fujiyoshi. Mix-Geneformer: Unified Representation Learning for Human and Mouse scRNA-seq Data. *arXiv preprint arXiv:2507.07454*, 2025. URL <https://arxiv.org/abs/2507.07454>.
- [155] Gyutaek Oh, Baekgyu Choi, Inkyung Jung, and Jong Chul Ye. scHyena: Foundation Model for Full-Length Single-Cell RNA-Seq Analysis in Brain. *arXiv preprint arXiv:2310.02713*, 2023. URL <https://arxiv.org/abs/2310.02713>.
- [156] Ofek Ophir, Orit Shefi, and Ofir Lindenbaum. Neuronal Cell Type Classification using Deep Learning. *arXiv preprint arXiv:2306.00528*, 2023. URL <https://arxiv.org/abs/2306.00528>.
- [157] Olga Ovcharenko, Florian Barkmann, Philip Toma, Imant Daunhawer, Julia Vogt, Sebastian Schelter, and Valentina Boeva. scSSL-Bench: Benchmarking Self-Supervised Learning for Single-Cell Data. *arXiv preprint arXiv:2506.10031*, 2025. URL <https://arxiv.org/abs/2506.10031>.
- [158] Steven Palayew, Bo Wang, and Gary Bader. Towards Applying Large Language Models to Complement Single-Cell Foundation Models. *arXiv preprint arXiv:2507.10039*, 2025. URL <https://arxiv.org/abs/2507.10039>.
- [159] Giovanni Palla, Sudarshan Babu, Payam Dibaeinia, James D. Pearce, Donghui Li, Aly A. Khan, Theofanis Karaletsos, and Jakub M. Tomczak. Scalable Single-Cell Gene Expression Generation with Latent Diffusion Models. *arXiv preprint arXiv:2511.02986*, 2025. URL <https://arxiv.org/abs/2511.02986>.
- [160] Shi Pan, Jianan Chen, and Maria Secrier. Teaching pathology foundation models to accurately predict gene expression with parameter efficient knowledge transfer. *arXiv preprint arXiv:2504.07061*, 2025. URL <https://arxiv.org/abs/2504.07061>.
- [161] Jisung Park, Seohyeon Kang, Daeun Yoo, Eunsu Lee, Seoin Cho, Wooyeop Choi, Ian Choi, James R. Evan, Daesoo Kim, Sonia Gandhi, and Minee L. Choi. Resolving superposition in AI for interpretability and cross-modal alignment in patient-neuronal images. *arXiv preprint arXiv:2606.31394*, 2026. URL <https://arxiv.org/abs/2606.31394>.
- [162] Yun Peng, Yujun Sun, Jia Ding, Bin Yan, Zhangyu Wang, Chunyang Wang, Chenyang Shu, Jian-Guo Zhou, and Shixiang Wang. oxo-call: Documentation-grounded Skill Augmentation for Accurate Bioinformatics Command-line Generation with Large Language Models. *arXiv preprint arXiv:2604.12387*, 2026. URL <https://arxiv.org/abs/2604.12387>.
- [163] Sai Teja Pusuluri, Alex H. Lang, Pankaj Mehta, and Horacio E. Castillo. Cellular reprogramming dynamics follow a simple one-dimensional reaction coordinate. *arXiv preprint arXiv:1505.03889*, 2015. URL <https://arxiv.org/abs/1505.03889>.
- [164] Cong Qi, Hanzhang Fang, Siqi Jiang, Xun Song, Tianxing Hu, and Wei Zhi. GeneMamba: An Efficient and Effective Foundation Model on Single Cell Data. *arXiv preprint arXiv:2504.16956*, 2025. URL <https://arxiv.org/abs/2504.16956>.
- [165] Jiaxin Qi, Yan Cui, Jinli Ou, Jianqiang Huang, and Gaogang Xie. Graph Neural Networks as a Substitute for Transformers in Single-Cell Transcriptomics. *arXiv preprint arXiv:2507.04125*, 2025. URL <https://arxiv.org/abs/2507.04125>.
- [166] Jiaxin Qi, Hang Li, Yan Cui, Yuhua Zheng, and Jianqiang Huang. Towards Universal Gene Regulatory Network Inference: Unlocking Generalizable Regulatory Knowledge in Single-cell Foundation Models. *arXiv preprint arXiv:2605.08128*, 2026. URL <https://arxiv.org/abs/2605.08128>.
- [167] Mufan Qiu, Xinyu Hu, Fengwei Zhan, Sukwon Yun, Jie Peng, Ruichen Zhang, Bhavya Kailkhura, Jiekun Yang, and Tianlong Chen. GRNFormer: A Biologically-Guided Framework for Integrating Gene Regulatory Networks into RNA Foundation Models. *arXiv preprint arXiv:2503.01682*, 2025. URL <https://arxiv.org/abs/2503.01682>.
- [168] Mufan Qiu, Genhui Zheng, Yinuo Xu, Ruichen Zhang, Ying Ding, Qi Long, and Tianlong Chen. Chreode: A Cell World Model for One-Step Temporal Dynamics and Perturbation Prediction. *arXiv preprint arXiv:2605.28111*, 2026. URL <https://arxiv.org/abs/2605.28111>.
- [169] Kalyan Ramakrishnan, Jonathan G. Hedley, Sisi Qu, Puneet K. Dokania, Philip H. S. Torr, Cesar A. Prada-Medina, Julien Fauqueur, and Kaspar Martens. Modeling Gene Expression Distributional Shifts for Unseen Genetic Perturbations. *arXiv preprint arXiv:2507.02980*, 2025. URL <https://arxiv.org/abs/2507.02980>.
- [170] Visweswaran Ravikumar, Aaresh Bhathena, Wajd N Al-Holou, Salar Fattahi, and Arvind Rao. Efficient inference of dynamic gene regulatory networks using discrete penalty.

- arXiv preprint arXiv:2507.23106*, 2025. URL <https://arxiv.org/abs/2507.23106>.
- [171] Ekaterina Redekop, Mara Pleasure, Zichen Wang, Kimberly Flores, Anthony Sisk, William Speier, and Corey W. Arnold. SPADE: Spatial Transcriptomics and Pathology Alignment Using a Mixture of Data Experts for an Expressive Latent Space. *arXiv preprint arXiv:2506.21857*, 2025. URL <https://arxiv.org/abs/2506.21857>.
- [172] Syed Asad Rizvi, Nazreen Pallikkavaliyaveetil, David Zhang, Zhuoyang Lyu, Nhi Nguyen, Hao-ran Lyu, Benjamin Christensen, Josue Ortega Caro, Antonio H. O. Fonseca, Emanuele Zappala, Maryam Bagherian, Christopher Averill, Chadi G. Abdallah, Amin Karbasi, Rex Ying, Maria Brbic, Rahul Madhav Dhodapkar, and David van Dijk. FIMP: Foundation Model-Informed Message Passing for Graph Neural Networks. *arXiv preprint arXiv:2210.09475*, 2022. URL <https://arxiv.org/abs/2210.09475>.
- [173] Joao F. Rocha, Ke Xu, Xingzhi Sun, Ananya Krishna, Dhananjay Bhaskar, Blanche Mongeon, Morgan Craig, Mark Gerstein, and Smita Krishnaswamy. STAGED: A Multi-Agent Neural Network for Learning Cellular Interaction Dynamics. *arXiv preprint arXiv:2507.11660*, 2025. URL <https://arxiv.org/abs/2507.11660>.
- [174] Selim Romero, Vignesh S. Kumar, Robert S. Chapkin, and James J. Cai. Quantum Generative Modeling of Single-Cell transcriptomes: Capturing Gene-Gene and Cell-Cell Interactions. *arXiv preprint arXiv:2510.12776*, 2025. URL <https://arxiv.org/abs/2510.12776>.
- [175] Till Rossner, Ziteng Li, Jonas Balke, Nikoo Salehfar, Tom Seifert, and Ming Tang. Integrating Single-Cell Foundation Models with Graph Neural Networks for Drug Response Prediction. *arXiv preprint arXiv:2504.14361*, 2025. URL <https://arxiv.org/abs/2504.14361>.
- [176] Jiafa Ruan, Ruijie Quan, Liyang Xu, Zongxin Yang, and Yi Yang. Beyond Independent Genes: Learning Module-Inductive Representations for Single-Cell Gene Perturbation Prediction. *arXiv preprint arXiv:2602.04901*, 2026. URL <https://arxiv.org/abs/2602.04901>.
- [177] Andrea Rubbi, Andrea Giuseppe Di Francesco, Mohammad Lotfollahi, and Pietro Lio. A Standardized Framework For Evaluating Gene Expression Generative Models. *arXiv preprint arXiv:2603.11244*, 2026. URL <https://arxiv.org/abs/2603.11244>.
- [178] Rabeya Tus Sadia, Md Atik Ahamed, and Qiang Cheng. CausalGeD: Blending Causality and Diffusion for Spatial Gene Expression Generation. *arXiv preprint arXiv:2502.07751*, 2025. URL <https://arxiv.org/abs/2502.07751>.
- [179] Manuel Schurch, Laura Boos, Viola Heinzelmann-Schwarz, Gabriele Gut, Michael Krauthammer, Andreas Wicki, and Tumor Profiler Consortium. Towards AI-Based Precision Oncology: A Machine Learning Framework for Personalized Counterfactual Treatment Suggestions based on Multi-Omics Data. *arXiv preprint arXiv:2402.12190*, 2024. URL <https://arxiv.org/abs/2402.12190>.
- [180] Arijit Sehanobish, Neal G. Ravindra, and David van Dijk. Gaining Insight into SARS-CoV-2 Infection and COVID-19 Severity Using Self-supervised Edge Features and Graph Neural Networks. *arXiv preprint arXiv:2006.12971*, 2020. URL <https://arxiv.org/abs/2006.12971>.
- [181] Peiting Shi, Ningfeng Que, Xianzhe Huang, Xiaofei Wang, and Jianzhong Jeff Xi. StateXDiff: Cell State-Contextualized Multimodal Diffusion for Single-Cell Perturbation Prediction. *arXiv preprint arXiv:2605.16104*, 2026. URL <https://arxiv.org/abs/2605.16104>.
- [182] Yaorui Shi, Jiaqi Yang, Changhao Nai, Sihang Li, Junfeng Fang, Xiang Wang, Zhiyuan Liu, and Yang Zhang. Language-Enhanced Representation Learning for Single-Cell Transcriptomics. *arXiv preprint arXiv:2503.09427*, 2025. URL <https://arxiv.org/abs/2503.09427>.
- [183] Zhiceng Shi, Changmiao Wang, Jun Wan, and Wenwen Min. Cross-Slice Knowledge Transfer via Masked Multi-Modal Heterogeneous Graph Contrastive Learning for Spatial Gene Expression Inference. *arXiv preprint arXiv:2603.22821*, 2026. URL <https://arxiv.org/abs/2603.22821>.
- [184] Junha Shin, Spencer Halberg-Spencer, Yuda Liu, Suvojit Hazra, Erika Da-Inn Lee, and Sushmita Roy. Modeling Dynamics, Cell Type Specificity, and Perturbations in Gene Regulatory Networks. *arXiv preprint arXiv:2602.18854*, 2026. URL <https://arxiv.org/abs/2602.18854>.
- [185] Mohammad Aamir Sohail, Ranga R. Sudharshan, S. Sandeep Pradhan, and Arvind Rao. Quantum Hamiltonian Learning using Time-Resolved Measurement Data and its Application to Gene Regulatory Network Inference. *arXiv preprint arXiv:2602.19496*, 2026. URL <https://arxiv.org/abs/2602.19496>.
- [186] Haoran Song, Jiarui Feng, Guangfu Li, Michael Province, Philip Payne, Yixin Chen, and Fuhai Li. Large Language Models Meet Graph Neural Networks for Text-Numeric Graph Reasoning. *arXiv preprint arXiv:2501.16361*, 2025. URL <https://arxiv.org/abs/2501.16361>.
- [187] Ran Song, Yinpu Bai, and Hui Liu. Fourier Asymmetric Attention on Domain Generalization

- for Pan-Cancer Drug Response Prediction. *arXiv preprint arXiv:2502.04034*, 2025. URL <https://arxiv.org/abs/2502.04034>.
- [188] Huan Souza and Pankaj Mehta. Parameter-free representations outperform single-cell foundation models on downstream benchmarks. *arXiv preprint arXiv:2602.16696*, 2026. URL <https://arxiv.org/abs/2602.16696>.
- [189] Yutong Sun, Sichen Zhu, and Peng Qiu. Integrating Pathology Foundation Models and Spatial Transcriptomics for Cellular Decomposition from Histology Images. *arXiv preprint arXiv:2507.07013*, 2025. URL <https://arxiv.org/abs/2507.07013>.
- [190] Maciej Sypetkowski, Joanna Krawczyk, ukasz Smolinski, Remigiusz Kinas, Przemyslaw Pietrzak, Tomasz Jetka, and Rafa Powalski. OmicsLM: A Multimodal Large Language Model for Multi-Sample Omics Reasoning. *arXiv preprint arXiv:2605.06728*, 2026. URL <https://arxiv.org/abs/2605.06728>.
- [191] Shawn Zheng Kai Tan, Aleix Puig-Barbe, Damien Goutte-Gattat, Caroline Eastwood, Brian Aeversmann, Alida Avola, James P Balhoff, Ismail Ugur Bayindir, Jasmine Belfiore, Anita Reane Caron, David S Fischer, Nancy George, Benjamin M Gyori, Melissa A Haendel, Charles Tapley Hoyt, Huseyin Kir, Tiago Lubiana, Nicolas Matentzoglou, James A Overton, Beverly Peng, et al. The Cell Ontology in the age of single-cell omics. *arXiv preprint arXiv:2506.10037*, 2025. URL <https://arxiv.org/abs/2506.10037>.
- [192] Junjie Tang, Changhu Wang, Feiyi Xiao, and Ruibin Xi. Single-cell gene regulatory network analysis for mixed cell populations with applications to COVID-19 single cell data. *arXiv preprint arXiv:2205.11274*, 2022. URL <https://arxiv.org/abs/2205.11274>.
- [193] Scienta Team, Ethan Bandasack, Vincent Bouget, Apolline Bruley, Yannis Cattan, Charlotte Claye, Matthew Corney, Julien Duquesne, Karim El Kanbi, Aziz Fouche, Pierre Marschall, and Francesco Strozzi. EVA: Towards a universal model of the immune system. *arXiv preprint arXiv:2602.10168*, 2026. URL <https://arxiv.org/abs/2602.10168>.
- [194] Binon Teji, Subhajit Bandyopadhyay, and Swarup Roy. Graph Attention Based Prioritization of Disease Responsible Genes from Multimodal Alzheimer’s Network. *arXiv preprint arXiv:2603.02273*, 2026. URL <https://arxiv.org/abs/2603.02273>.
- [195] Panagiotis Tigas, Yashas Annadani, Andrew Jesson, Bernhard Scholkopf, Yarin Gal, and Stefan Bauer. Interventions, Where and How? Experimental Design for Causal Models at Scale. *arXiv preprint arXiv:2203.02016*, 2022. URL <https://arxiv.org/abs/2203.02016>.
- [196] Tsz Pan Tong, Aoran Wang, George Panagopoulos, and Jun Pang. Integrating Optimal Transport and Structural Inference Models for GRN Inference from Single-cell Data. *arXiv preprint arXiv:2409.15080*, 2024. URL <https://arxiv.org/abs/2409.15080>.
- [197] Ching-Huei Tsou, Michal Ozery-Flato, Ella Barkan, Diwakar Mahajan, and Ben Shapira. BioVERSE: Representation Alignment of Biomedical Modalities to LLMs for Multi-Modal Reasoning. *arXiv preprint arXiv:2510.01428*, 2025. URL <https://arxiv.org/abs/2510.01428>.
- [198] Alistair Turcan, Kexin Huang, Lei Li, and Martin Jinye Zhang. TusoAI: Agentic Optimization for Scientific Methods. *arXiv preprint arXiv:2509.23986*, 2025. URL <https://arxiv.org/abs/2509.23986>.
- [199] Muhammad Umar, Andras Lakatos, Muhammad Asif, and Arif Mahmood. Lower-dimensional projections of cellular expression improves cell type classification from single-cell RNA sequencing. *arXiv preprint arXiv:2410.09964*, 2024. URL <https://arxiv.org/abs/2410.09964>.
- [200] Abicumaran Uthamacumaran and Hector Zenil. A Review of Mathematical and Computational Methods in Cancer Dynamics. *arXiv preprint arXiv:2201.02055*, 2022. URL <https://arxiv.org/abs/2201.02055>.
- [201] Abicumaran Uthamacumaran and Hector Zenil. Neurosymbolic Learning for Predicting Cell Fate Decisions from Longitudinal Single Cell Transcriptomics in Paediatric Acute Myeloid Leukemia. *arXiv preprint arXiv:2508.13199*, 2025. URL <https://arxiv.org/abs/2508.13199>.
- [202] Pritika Vig, Ren-Chin Wu, and William Lotter. Do Pathology Foundation Models Encode Disease Progression? A Pseudotime Analysis of Visual Representations. *arXiv preprint arXiv:2601.21334*, 2026. URL <https://arxiv.org/abs/2601.21334>.
- [203] Haoran Wang, Xuanyi Zhang, Shuangfang Fang, Longke Ran, Ziqing Deng, Yong Zhang, Yuxiang Li, and Shaoshuai Li. Open World Knowledge Aided Single-Cell Foundation Model with Robust Cross-Modal Cell-Language Pre-training. *arXiv preprint arXiv:2601.05648*, 2026. URL <https://arxiv.org/abs/2601.05648>.
- [204] Hongxiao Wang, Yang Yang, Zhuo Zhao, Pengfei Gu, Nishchal Sapkota, and Danny Z. Chen. PathGPTomic: A Balanced Multi-modal Learning Framework for Survival Outcome Prediction. *arXiv preprint arXiv:2403.11375*, 2024. URL <https://arxiv.org/abs/2403.11375>.

- [205] Mingxuan Wang, Cheng Chen, Gaoyang Jiang, Zijia Ren, Chuangxin Zhao, Lu Shi, and Yanbiao Ma. ScDiVa: Masked Discrete Diffusion for Joint Modeling of Single-Cell Identity and Expression. *arXiv preprint arXiv:2602.03477*, 2026. URL <https://arxiv.org/abs/2602.03477>.
- [206] Qing Wang, Yining Pan, Minghao Zhou, Zijia Tang, Yanfei Wang, Guangyu Wang, and Qianqian Song. scDrugMap: Benchmarking Large Foundation Models for Drug Response Prediction. *arXiv preprint arXiv:2505.05612*, 2025. URL <https://arxiv.org/abs/2505.05612>.
- [207] Weikang Wang, Dante Poe, Ke Ni, and Jianhua Xing. Cell phenotypic transition proceeds through concerted reorganization of gene regulatory network. *arXiv preprint arXiv:2107.03581*, 2021. URL <https://arxiv.org/abs/2107.03581>.
- [208] Yue Wang, Peng Zheng, Yu-Chen Cheng, Zikun Wang, and Aleksandr Aravkin. WENDY: Covariance Dynamics Based Gene Regulatory Network Inference. *arXiv preprint arXiv:2407.00754*, 2024. URL <https://arxiv.org/abs/2407.00754>.
- [209] Yuechen Wang, Lina Jia, Qinglong Wang, and Feng Tian. ORBIT: Learning Gene Program Co-Activation Structure for Cell-Type-Stratified Pathway Rewiring Analysis in Single-Cell Transcriptomics. *arXiv preprint arXiv:2605.02142*, 2026. URL <https://arxiv.org/abs/2605.02142>.
- [210] Yuhao Wang, Liam Solus, Karren Dai Yang, and Caroline Uhler. Permutation-based Causal Inference Algorithms with Interventions. *arXiv preprint arXiv:1705.10220*, 2017. URL <https://arxiv.org/abs/1705.10220>.
- [211] Zhen Wang, Yiming Gao, Jieyuan Liu, Enze Ma, Jefferson Chen, Mark Antkowiak, Mengzhou Hu, JungHo Kong, Dexter Pratt, Zhiting Hu, Wei Wang, Trey Ideker, and Eric P. Xing. CellMaster: Collaborative Cell Type Annotation in Single-Cell Analysis. *arXiv preprint arXiv:2602.13346*, 2026. URL <https://arxiv.org/abs/2602.13346>.
- [212] Zhenyu Wang, Zikang Wang, Jiyue Jiang, Peng Chen, Xiangyu Shi, and Yu Li. Large Language Models in Bioinformatics: A Survey. *arXiv preprint arXiv:2503.04490*, 2025. URL <https://arxiv.org/abs/2503.04490>.
- [213] Zhiruo Wang, Haoyu Dong, Ran Jia, Jia Li, Zhiyi Fu, Shi Han, and Dongmei Zhang. TUTA: Tree-based Transformers for Generally Structured Table Pre-training. *arXiv preprint arXiv:2010.12537*, 2020. URL <https://arxiv.org/abs/2010.12537>.
- [214] Sachini Weerasekara, Natasha Darras, Sagar Kamarthi, Colles Price, and Jacqueline Isaacs. Prototype Guided Post-pretraining for Single-Cell Representation Learning. *arXiv preprint arXiv:2605.07938*, 2026. URL <https://arxiv.org/abs/2605.07938>.
- [215] Zhijian Wei, Runze Ma, Zichen Wang, Zhongmin Li, Shuotong Song, and Shuangjia Zheng. VC-World: A Biological World Model for Virtual Cell Simulation. *arXiv preprint arXiv:2512.00306*, 2025. URL <https://arxiv.org/abs/2512.00306>.
- [216] Ethan Weinberger, Ryan Conrad, and Tal Ashuach. Modeling variable guide efficiency in pooled CRISPR screens with ContrastiveVI+. *arXiv preprint arXiv:2411.08072*, 2024. URL <https://arxiv.org/abs/2411.08072>.
- [217] Chaochen Wu, Meiyun Zuo, and Lei Xie. eDOC: Explainable Decoding Out-of-domain Cell Types with Evidential Learning. *arXiv preprint arXiv:2411.00054*, 2024. URL <https://arxiv.org/abs/2411.00054>.
- [218] Dongxia Wu, Mingyu Li, Yuhui Zhang, Anurendra Kumar, Emma Lundberg, Serena Yeung-Levy, and Emily B. Fox. PerturbCellRL: Verifier-Guided Reinforcement Learning for Single-Cell Perturbation Prediction. *arXiv preprint arXiv:2606.27752*, 2026. URL <https://arxiv.org/abs/2606.27752>.
- [219] Jinke Wu, Yifan Wang, Siyu Yi, Caiyang Yu, Ziyue Qiao, Nan Yin, Jiancheng Lv, and Wei Ju. scGTN: Deep Siamese Graph Transformer Network for Single-cell RNA Sequencing Clustering. *arXiv preprint arXiv:2606.18672*, 2026. URL <https://arxiv.org/abs/2606.18672>.
- [220] Menghua Wu, Yujia Bao, Regina Barzilay, and Tommi Jaakkola. Sample, estimate, aggregate: A recipe for causal discovery foundation models. *arXiv preprint arXiv:2402.01929*, 2024. URL <https://arxiv.org/abs/2402.01929>.
- [221] Weiyi Wu, Xinwen Xu, Xingjian Diao, Siting Li, Zhi Wei, Alma Andersson, and Jiang Gui. HiST: A Hierarchical Sparse Transformer for Cross-Modal Spatial Transcriptomics Modeling. *arXiv preprint arXiv:2606.14251*, 2026. URL <https://arxiv.org/abs/2606.14251>.
- [222] Rijie Xi, Weikang Xu, Wei Xiong, Yuannong Ye, and Bin Zhao. Gene regulatory network inference algorithm based on spectral signed directed graph convolution. *arXiv preprint arXiv:2512.11927*, 2025. URL <https://arxiv.org/abs/2512.11927>.
- [223] Jinxi Xiang, Siyu Hou, Yuchen Li, Ryan Quinton, Xiaoming Zhang, Feyisope Eweje, Xiangde Luo, Yijiang Chen, Zhe Li, Colin Bergstrom, Ted Kim, Sierra Willens, Francesca Maria Olguin, Matthew Abikenari, Andrew Heider, Sanjeeth Rajaram, Joel Neal, Maximilian Diehn, Xiang Zhou, and Ruijiang Li. A Multimodal Foundation Model of Spatial Transcriptomics and Histology for Biological Discovery and Clinical Prediction. *arXiv preprint*

- arXiv:2604.03630*, 2026. URL <https://arxiv.org/abs/2604.03630>.
- [224] Feiyi Xiao, Junjie Tang, Huaying Fang, and Ruibin Xi. Gene regulatory network in single cells based on the Poisson log-normal model. *arXiv preprint arXiv:2111.04037*, 2021. URL <https://arxiv.org/abs/2111.04037>.
- [225] Yucheng Xing, Pei Liu, Jingying Ma, Ruping Hong, Jiangdong Qiu, Tianyu Liu, Kai He, Ling Huang, and Mengling Feng. Bridging the Modality Bottleneck in Pathology MIL through Virtual Molecular Staining. *arXiv preprint arXiv:2605.16392*, 2026. URL <https://arxiv.org/abs/2605.16392>.
- [226] Ping Xu, Zhiyuan Ning, Meng Xiao, Guihai Feng, Xin Li, Yuanchun Zhou, and Pengfei Wang. scCDCG: Efficient Deep Structural Clustering for single-cell RNA-seq via Deep Cut-informed Graph Embedding. *arXiv preprint arXiv:2404.06167*, 2024. URL <https://arxiv.org/abs/2404.06167>.
- [227] Ping Xu, Zhiyuan Ning, Pengjiang Li, Wenhao Liu, Pengyang Wang, Jiaxu Cui, Yuanchun Zhou, and Pengfei Wang. scSiameseClu: A Siamese Clustering Framework for Interpreting single-cell RNA Sequencing Data. *arXiv preprint arXiv:2505.12626*, 2025. URL <https://arxiv.org/abs/2505.12626>.
- [228] Ping Xu, Pengfei Wang, Zhiyuan Ning, Meng Xiao, Min Wu, and Yuanchun Zhou. Soft Graph Clustering for single-cell RNA Sequencing Data. *arXiv preprint arXiv:2507.09890*, 2025. URL <https://arxiv.org/abs/2507.09890>.
- [229] Ping Xu, Zaitian Wang, Zhirui Wang, Pengjiang Li, Jiajia Wang, Ran Zhang, Pengfei Wang, and Yuanchun Zhou. scCluBench: Comprehensive Benchmarking of Clustering Algorithms for Single-Cell RNA Sequencing. *arXiv preprint arXiv:2512.02471*, 2025. URL <https://arxiv.org/abs/2512.02471>.
- [230] Ping Xu, Zaitian Wang, Zhirui Wang, Pengjiang Li, Ran Zhang, Gaoyang Li, Hanyu Xie, Jiajia Wang, Yuanchun Zhou, and Pengfei Wang. scUnified: An AI-Ready Standardized Resource for Single-Cell RNA Sequencing Analysis. *arXiv preprint arXiv:2509.25884*, 2025. URL <https://arxiv.org/abs/2509.25884>.
- [231] Ping Xu, Pengjiang Li, Tian Du, Zaitian Wang, Jiawei Gu, Zhiyuan Ning, Ziyue Qiao, Pengfei Wang, and Yuanchun Zhou. scLLM-DSC: LLM-Knowledge Enhanced Cross-Modal Deep Structural Clustering for Single-Cell RNA Sequencing. *arXiv preprint arXiv:2606.13007*, 2026. URL <https://arxiv.org/abs/2606.13007>.
- [232] Tianyang Xu, Tianci Liu, Niraj Rayamajhi, Ryan Patrick, Kranthi Varala, Ying Li, and Jing Gao. Prior-Guided Multi-Omic Transformers for Single-Cell Gene Regulatory Network Inference. *arXiv preprint arXiv:2606.00685*, 2026. URL <https://arxiv.org/abs/2606.00685>.
- [233] Yinuo Xu, Yan Cui, Mingyao Li, and Zhi Huang. Adaptive Multi-Scale Integration Unlocks Robust Cell Annotation in Histopathology Images. *arXiv preprint arXiv:2511.13586*, 2025. URL <https://arxiv.org/abs/2511.13586>.
- [234] Albert Xue, Jingyou Rao, Sriram Sankararaman, and Harold Pimentel. dotears: Scalable, consistent DAG estimation using observational and interventional data. *arXiv preprint arXiv:2305.19215*, 2023. URL <https://arxiv.org/abs/2305.19215>.
- [235] Shuailin Xue, Fangfang Zhu, Changmiao Wang, and Wenwen Min. stEnTrans: Transformer-based deep learning for spatial transcriptomics enhancement. *arXiv preprint arXiv:2407.08224*, 2024. URL <https://arxiv.org/abs/2407.08224>.
- [236] Maria Yampolskaya, Laertis Ikononou, and Pankaj Mehta. Collective gene dynamics leave signatures of decision landscapes in cell fate coordinates. *arXiv preprint arXiv:2506.04219*, 2025. URL <https://arxiv.org/abs/2506.04219>.
- [237] Shuai Yan, Qingzhi Yu, Wengfeng Dai, and Xiang Cheng. GP-DHT: A Dual-Head Transformer with Contrastive Learning for Predicting Gene Regulatory Relationships across Species from Single-Cell Data. *arXiv preprint arXiv:2601.10995*, 2026. URL <https://arxiv.org/abs/2601.10995>.
- [238] Zichao Yan, Yan Wu, Mica Xu Ji, Chaitra Agrahar, Esther Wershof, Marcel Nassar, Mehrshad Sadria, Ridvan Eksi, Vladimir Trifonov, Ignacio Ibarra, Telmo Felgueira, Bazej Osinski, and Rory Stark. PRiMeFlow: Capturing Complex Expression Heterogeneity in Perturbation Response Modelling. *arXiv preprint arXiv:2604.13986*, 2026. URL <https://arxiv.org/abs/2604.13986>.
- [239] Changchun Yang, Haoyang Li, Yushuai Wu, Yilan Zhang, Yifeng Jiao, Yu Zhang, Rihan Huang, Yuan Cheng, Yuan Qi, Xin Guo, and Xin Gao. PAST: A multimodal single-cell foundation model for histopathology and spatial transcriptomics in cancer. *arXiv preprint arXiv:2507.06418*, 2025. URL <https://arxiv.org/abs/2507.06418>.
- [240] Rui Yang, Wenrui Dai, Chenglin Li, Junni Zou, Dapeng Wu, and Hongkai Xiong. scBiGNN: Bilevel Graph Representation Learning for Cell Type Classification from Single-cell RNA Sequencing Data. *arXiv preprint arXiv:2312.10310*, 2023. URL <https://arxiv.org/abs/2312.10310>.
- [241] Yan Yang, Md Zakir Hossain, Eric A Stone, and Shafin Rahman. Exemplar Guided Deep Neural Network for Spatial Transcriptomics Analysis of Gene Expression Prediction. *arXiv preprint arXiv:2210.16721*, 2022. URL <https://arxiv.org/abs/2210.16721>.

- [242] Yan Yang, Md Zakir Hossain, Xuesong Li, Shafin Rahman, and Eric Stone. Spatial Transcriptomics Analysis of Zero-shot Gene Expression Prediction. *arXiv preprint arXiv:2401.14772*, 2024. URL <https://arxiv.org/abs/2401.14772>.
- [243] Yehui Yang, Zelin Zang, Xienan Zheng, Yuzhe Jia, Changxi Chi, Jingbo Zhou, Chang Yu, Jinlin Wu, Fuji Yang, Jiebo Luo, Zhen Lei, and Stan Z. Li. MAT-Cell: A Multi-Agent Tree-Structured Reasoning Framework for Batch-Level Single-Cell Annotation. *arXiv preprint arXiv:2604.06269*, 2026. URL <https://arxiv.org/abs/2604.06269>.
- [244] Yongjian Yang. Deciphering Cell Systems: Machine Learning Perspectives And Approaches For The Analysis Of Single-Cell Data. *arXiv preprint arXiv:2409.19482*, 2024. URL <https://arxiv.org/abs/2409.19482>.
- [245] Wenjin Ye, Yuanchen Ma, Junkai Xiang, Hongjie Liang, Tao Wang, Qiuling Xiang, Andy Peng Xiang, Wu Song, Weiqiang Li, and Weijun Huang. Objectively Evaluating the Reliability of Cell Type Annotation Using LLM-Based Strategies. *arXiv preprint arXiv:2409.15678*, 2024. URL <https://arxiv.org/abs/2409.15678>.
- [246] Chenglei Yu, Chuanrui Wang, Bangyan Liao, and Tailin Wu. scDFM: Distributional Flow Matching Model for Robust Single-Cell Perturbation Prediction. *arXiv preprint arXiv:2602.07103*, 2026. URL <https://arxiv.org/abs/2602.07103>.
- [247] Zhen Yu, Jianan Han, Yang Liu, and Qingchao Chen. scDD: Latent Codes Based scRNA-seq Dataset Distillation with Foundation Model Knowledge. *arXiv preprint arXiv:2503.04357*, 2025. URL <https://arxiv.org/abs/2503.04357>.
- [248] Xinyu Yuan, Zhihao Zhan, Zuobai Zhang, Manqi Zhou, Jianan Zhao, Boyu Han, Yue Li, and Jian Tang. Cell-ontology guided transcriptome foundation model. *arXiv preprint arXiv:2408.12373*, 2024. URL <https://arxiv.org/abs/2408.12373>.
- [249] Xinyu Yuan, Xixian Liu, Ya Shi Zhang, Zuobai Zhang, Hongyu Guo, and Jian Tang. PerturbDiff: Functional Diffusion for Single-Cell Perturbation Modeling. *arXiv preprint arXiv:2602.19685*, 2026. URL <https://arxiv.org/abs/2602.19685>.
- [250] Ying Yuan, Xing-Yue Monica Ge, Aaron Archer Waterman, Tommaso Biancalani, David Richmond, Yogesh Pandit, Avtar Singh, Russell Littman, Jin Liu, Jan-Christian Huetter, and Vladimir Ermakov. HypoGeneAgent: A Hypothesis Language Agent for Gene-Set Cluster Resolution Selection Using Perturb-seq Datasets. *arXiv preprint arXiv:2509.09740*, 2025. URL <https://arxiv.org/abs/2509.09740>.
- [251] Zhen Yuan, Shaoqing Jiao, Yihang Xiao, and Jiajie Peng. scMamba: A Scalable Foundation Model for Single-Cell Multi-Omics Integration Beyond Highly Variable Feature Selection. *arXiv preprint arXiv:2506.20697*, 2025. URL <https://arxiv.org/abs/2506.20697>.
- [252] Sayeem Bin Zaman, Fahim Hafiz, and Riasat Azim. SpatialMAGIC: A Hybrid Framework Integrating Graph Diffusion and Spatial Attention for Spatial Transcriptomics Imputation. *arXiv preprint arXiv:2603.06780*, 2026. URL <https://arxiv.org/abs/2603.06780>.
- [253] Mehrdad Zandigohar and Yang Dai. Information retrieval in single cell chromatin analysis using TF-IDF transformation methods. *arXiv preprint arXiv:2212.05184*, 2022. URL <https://arxiv.org/abs/2212.05184>.
- [254] Zehua Zeng and Hongwu Du. Revolutionizing Single Cell Analysis: The Power of Large Language Models for Cell Type Annotation. *arXiv preprint arXiv:2304.02697*, 2023. URL <https://arxiv.org/abs/2304.02697>.
- [255] Xianghao Zhan, Jingyu Xu, Yuanning Zheng, Zinaida Good, and Olivier Gevaert. SAGE-FM: A lightweight and interpretable spatial transcriptomics foundation model. *arXiv preprint arXiv:2601.15504*, 2026. URL <https://arxiv.org/abs/2601.15504>.
- [256] Fan Zhang, Tianyu Liu, Zhihong Zhu, Hao Wu, Haixin Wang, Donghao Zhou, Yefeng Zheng, Kun Wang, Xian Wu, and Pheng-Ann Heng. CellVerse: Do Large Language Models Really Understand Cell Biology? *arXiv preprint arXiv:2505.07865*, 2025. URL <https://arxiv.org/abs/2505.07865>.
- [257] Han Zhang, Guo-Hua Yuan, Chaohao Yuan, Tingyang Xu, Tian Bian, Hong Cheng, Wenbing Huang, Deli Zhao, and Yu Rong. Lingshu-Cell: A generative cellular world model for transcriptome modeling toward virtual cells. *arXiv preprint arXiv:2603.25240*, 2026. URL <https://arxiv.org/abs/2603.25240>.
- [258] Heming Zhang, Tim Xu, Dekang Cao, Shunning Liang, Guntaas Shergill, Nicholas Hadas, Lars Schimmelpfennig, Levi Kaster, Di Huang, Guangfu Li, S. Peter Goedgebuure, David DeNardo, Li Ding, Ryan C. Fields, J Philip Miller, Pirooz Eghtesady, Carlos Cruchaga, William Buchser, Jonathan Cooper, Marco Sardiello, et al. OmniCellTOSG: The First Cell Text-Omic Signaling Graphs Dataset for Graph Language Foundation Modeling. *arXiv preprint arXiv:2504.02148*, 2025. URL <https://arxiv.org/abs/2504.02148>.
- [259] Luxuan Zhang, Douglas Jiang, Qinglong Wang, Haoqi Sun, and Feng Tian. Contrastive Learning Enhances Language Model Based Cell Embeddings for Low-Sample Single Cell Transcriptomics.

- arXiv preprint arXiv:2509.23543*, 2025. URL <https://arxiv.org/abs/2509.23543>.
- [260] Rongyu Zhang, Hongyu Dong, Gaole Dai, Ziqi Qiao, Shenli Zheng, Yuan Zhang, Aosong Cheng, Xiaowei Chi, Jincai Luo, Pin Li, Li Du, Dan Wang, Yuan Du, Xudong Xing, Jianxu Chen, and Shanghang Zhang. Key-Embedded Privacy for Decentralized AI in Biomedical Omics. *arXiv preprint arXiv:2603.28334*, 2026. URL <https://arxiv.org/abs/2603.28334>.
- [261] Wei Zhang, Xun Jiang, Yuesi Xi, and Ming Tang. CisTransCell: Single-Cell Perturbation Prediction via Gene Function, Regulatory Control, and Cellular Context. *arXiv preprint arXiv:2606.13713*, 2026. URL <https://arxiv.org/abs/2606.13713>.
- [262] Yiyang Zhang, Bokai Zhao, Xiaoru Zhang, Zongchang Du, Xiaojuan Sun, and Tianzi Jiang. SpCAST: Decoding spatial transcriptomics at single-cell resolution with fast and interpretable analysis. *arXiv preprint arXiv:2605.26904*, 2026. URL <https://arxiv.org/abs/2605.26904>.
- [263] Zhonghui Zhang, Feng Jiang, Shaowei Qin, Jiahao Zhao, and Min Yang. GATHER: Convergence-Centric Hyper-Entity Retrieval for Zero-Shot Cell-Type Annotation. *arXiv preprint arXiv:2605.06403*, 2026. URL <https://arxiv.org/abs/2605.06403>.
- [264] Dingping Zhao, Jie Lin, Feng Xu, and Zhengwei Xie. Modeling Cell-Cycle-Aware Single-Cell Drug Perturbation Responses. *arXiv preprint arXiv:2606.30695*, 2026. URL <https://arxiv.org/abs/2606.30695>.
- [265] Huan Zhao, Yiming Liu, Jina Yao, Ling Xiong, Zexin Zhou, and Zixing Zhang. Celler: A Genomic Language Model for Long-Tailed Single-Cell Annotation. *arXiv preprint arXiv:2504.00020*, 2025. URL <https://arxiv.org/abs/2504.00020>.
- [266] Jiahao Zhao, Feng Jiang, Shaowei Qin, Zhonghui Zhang, Junhao Liu, Guibing Guo, Hamid Alinejad-Rokny, and Min Yang. SC-Arena: A Natural Language Benchmark for Single-Cell Reasoning with Knowledge-Augmented Evaluation. *arXiv preprint arXiv:2602.23199*, 2026. URL <https://arxiv.org/abs/2602.23199>.
- [267] Suyuan Zhao, Jiahuan Zhang, and Zaiqing Nie. Large-Scale Cell Representation Learning via Divide-and-Conquer Contrastive Learning. *arXiv preprint arXiv:2306.04371*, 2023. URL <https://arxiv.org/abs/2306.04371>.
- [268] Suyuan Zhao, Yizhen Luo, Ganbo Yang, Yan Zhong, Hao Zhou, and Zaiqing Nie. SToFM: a Multi-scale Foundation Model for Spatial Transcriptomics. *arXiv preprint arXiv:2507.11588*, 2025. URL <https://arxiv.org/abs/2507.11588>.
- [269] Wenmin Zhao, Ana Solaguren-Beascoa, Grant Neilson, Louwai Muhammed, Liisi Laaniste, and Sera Aylin Cakiroglu. Multi-modal single-cell foundation models via dynamic token adaptation. *arXiv preprint arXiv:2504.13049*, 2025. URL <https://arxiv.org/abs/2504.13049>.
- [270] Zhiwei Zheng and Kevin Bryson. TwinPurify: Purifying gene expression data to reveal tumor-intrinsic transcriptional programs via self-supervised learning. *arXiv preprint arXiv:2601.18640*, 2026. URL <https://arxiv.org/abs/2601.18640>.
- [271] Juexiao Zhou, Bin Zhang, Xiuying Chen, Haoyang Li, Xiaopeng Xu, Siyuan Chen, and Xin Gao. Automated Bioinformatics Analysis via AutoBA. *arXiv preprint arXiv:2309.03242*, 2023. URL <https://arxiv.org/abs/2309.03242>.
- [272] Qifeng Zhou, Lei Yu, Yuzhi Guo, Yuwei Miao, Hehuan Ma, Wenliang Zhong, Lin Xu, and Junzhou Huang. scpFormer: A Foundation Model for Unified Representation and Integration of the Single-Cell Proteomics. *arXiv preprint arXiv:2604.20003*, 2026. URL <https://arxiv.org/abs/2604.20003>.
- [273] Junchao Zhu, Ruining Deng, Junlin Guo, Tianyuan Yao, Chongyu Qu, Juming Xiong, Siqi Lu, Zhengyi Lu, Yanfan Zhu, Marilyn Lions, Yuechen Yang, Yalin Zheng, Yu Wang, Shilin Zhao, Haichun Yang, and Yuankai Huo. SCR2-ST: Combine Single Cell with Spatial Transcriptomics for Efficient Active Sampling via Reinforcement Learning. *arXiv preprint arXiv:2512.13635*, 2025. URL <https://arxiv.org/abs/2512.13635>.
- [274] Junchao Zhu, Ruining Deng, Tianyuan Yao, Juming Xiong, Chongyu Qu, Junlin Guo, Siqi Lu, Yucheng Tang, Daguang Xu, Mengmeng Yin, Yu Wang, Shilin Zhao, Yaohong Wang, Haichun Yang, and Yuankai Huo. MagNet: Multi-Level Attention Graph Network for Predicting High-Resolution Spatial Transcriptomics. *arXiv preprint arXiv:2502.21011*, 2025. URL <https://arxiv.org/abs/2502.21011>.
- [275] Junhao Zhu, Kevin Zhang, Dehan Kong, and Zhaolei Zhang. LLOT: application of Laplacian Linear Optimal Transport in spatial transcriptome reconstruction. *arXiv preprint arXiv:2408.15149*, 2024. URL <https://arxiv.org/abs/2408.15149>.
- [276] Yishun Zhu, Jiaxin Qi, Jian Wang, Yuhua Zheng, and Jianqiang Huang. Spatial Transcriptomics as Images for Large-Scale Pretraining. *arXiv preprint arXiv:2603.13432*, 2026. URL <https://arxiv.org/abs/2603.13432>.